Availability-Constrained Factor Discovery for 30-Minute Glucose Forecasting on CGMacros

Joshua Lu

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Abstract

Glucose monitors are increasingly combined with food logs and activity trackers, but it remains unclear what lifestyle records add to short-term glucose forecasting beyond glucose history. Using CGMacros data from 45 free-living adults, we investigate 30-minute-ahead glucose forecasting from a 120-minute history, evaluating both later periods of participants seen in training and previously unseen participants. We reimplement two established baselines, the GlucoBench linear regression and a local autoregressive model, and examine whether additional factors improve a fixed baseline. Candidate factors are constructed manually, through a theory-constrained formula grammar based on meal absorption, using OpenFE, and through a search guided by a large language model (LLM). The LLM-guided search is compared with random search matched on the number of unique valid candidate evaluations, using validation data only. Each factor block is fitted to the baseline residual. GlucoBench was the strongest of six main comparison models, although exploratory recurrent and convolutional networks achieved lower test errors within subjects. Improvements from additional factors were small: five of eight primary comparisons had absolute RMSE changes below 0.1 mg/dL. For unseen participants, manual and OpenFE factors reduced RMSE by 0.245 and 0.244 mg/dL, respectively, whereas manual factors showed no clear improvement within subjects. LLM guidance did not outperform random search on validation data. Two exploratory validation-only follow-up analyses found small improvements when the selected factor blocks were added to fixed GRU and TCN forecasters, but no clear overall gain from laboratory-state interactions. With only nine unseen-subject test participants and a previously used test split, the factor comparisons are descriptive rather than independently confirmatory; estimated gains were also sensitive to cohort definition and assumptions about meal-log availability.

1 Introduction

Continuous glucose monitors (CGMs) are now routinely combined with food logs and activity trackers; Mortazavi and Gutierrez-Osuna [15] review the digital tools used for diet monitoring. A natural question is whether such lifestyle records help to predict glucose over the next half hour beyond what the glucose history already contains. CGMacros [8], to its authors' knowledge the first publicly available database that combines CGM, meal macronutrients, food photographs, activity-tracker data and health parameters, allows this question to be asked for 45 adults with and without diabetes.

The first task of this project is to define the prediction problem and to establish what existing methods achieve. We predict the glucose value 30 minutes ahead from the preceding 120 minutes of released CGM data and compare two settings: forecasting later periods of participants whose earlier data are available for fitting, and forecasting for participants who were never seen in training. Established methods are reimplemented as baselines (Section 5.1 describes the adaptations): the linear regression of the GlucoBench benchmark, where simple linear and ARIMA models were hard to beat and the benefit of covariates depended on the data set [18], and a first-order autoregressive model with a time-varying coefficient [21]. Persistence, a linear trend, and ridge and gradient-boosted models on engineered glucose features serve as references.

The second task is to identify additional factors in the data set and to incorporate them so that prediction improves. The evidence that exogenous inputs help is mixed. In type 1 diabetes, a support vector regression given glucose and daily events did not beat a glucose-only ARIMA model until the events were turned into features by a physiological model of meal absorption [3]. Such factors are often engineered by hand [2].

Automated feature generation, as in OpenFE [25], and large language models (LLMs) that write feature code or programs and are guided by validation scores [10, 1] are promising alternatives, but these methods were developed for tables of independent rows with random splits. CGM data are repeated measures from few participants, and every candidate factor must be computable at the time of the forecast.

We therefore hold an adapted published forecaster fixed and ask what a candidate factor block adds to it: each block is fitted on the residual of the GlucoBench regression, and the reported quantity is the paired change in subject-equal RMSE under both protocols. Candidates come from four sources: manually engineered diet, activity and participant features (M); a small formula grammar that respects prediction-time availability and meal-absorption physiology (T); OpenFE restricted to causal, row-wise operators (O); and an LLM-guided search, compared with a random search stopped at the same number of unique valid fitness evaluations and judged on validation data only (L and R). Screening, selection and tuning use training and validation data; the selected factors and the planned comparisons are then fixed, and the formal comparisons are then run once on a test split that had already been used in earlier project stages (Section 6.4).

Three questions structure the study:

- **RQ1**: Do diet, activity and baseline-metabolic features improve the fixed baseline, and are automatically constructed factors better than manual ones?
- **RQ2**: Does baseline metabolic state help more when it enters as meal-by-state interactions than as additive features (the laboratory-state interactions were not part of the formal test comparisons and were evaluated afterwards on validation data only; Section 7.8.2)?
- **RQ3**: How do theory-constrained construction, OpenFE and LLM-guided search compare in gain, parsimony and interpretability?

A validity check asks whether the interpolated glucose history changes the conclusions about diet. A hypothesis and its refutation condition were set for each question before the factor experiments were run and are stated with the experimental design (Section 6).

The main contributions of this study are:

- A 30-minute glucose forecasting task on CGMacros, with reimplemented baselines evaluated for both within-subject and unseen-subject prediction.
- A factor discovery framework that constructs manual, theory-constrained, automated and LLM-guided candidate features under common prediction-time availability rules, and integrates selected factors through residual augmentation of a fixed baseline.
- A systematic comparison of the predictive gains and stability of the constructed factors, including negative findings and analyses of sensitivity to evaluation protocol, cohort definition, meal-log timing and glucose-history representation. The LLM-guided search is compared with random search on validation data only.

Two additional exploratory validation-only analyses examine the transfer of selected factor blocks to GRU and TCN forecasters and the predictive value of laboratory-state interactions.

2 Related Work

Two bodies of work are relevant: glucose forecasting with and without lifestyle data, which frames the task and its baselines, and methods that construct and screen candidate features, including LLM-based ones, which frame the search for additional factors.

2.1 Glucose forecasting with lifestyle data

Sparacino et al. [21] re-estimated a first-order autoregressive model with a time-varying coefficient at every time point to predict glucose up to 30 minutes ahead in 28 volunteers with type 1 diabetes. GlucoBench [18] compares forecasters on five public CGM data sets: chronological training, validation and in-distribution test

periods for 90% of the subjects, and an out-of-distribution test on the remaining subjects. Linear regression and ARIMA were among the most accurate models, deep models generalized better to held-out subjects, and the effect of covariates varied with the data set (mixed on the larger sets, lower accuracy on two small ones, which the authors attribute to overfitting). A linear model of the glucose history is therefore a strong reference against which every factor must be judged, and results should be reported both for later periods of known participants and for unseen participants. We adopt the GlucoBench linear regression as the fixed base and the autoregressive model as a second adapted baseline.

Whether exogenous inputs help is less settled. Bunescu et al. [3] report that a support vector regression given glucose and daily events did no better than an ARIMA model on glucose alone; when the events were first passed through a physiological model of meal absorption, insulin and glucose dynamics, it improved on ARIMA (30-minute RMSE 19.6 against 22.9 mg/dL), with per-patient models and 200 hand-selected evaluation points from five patients. The authors attribute the earlier failure to inaccurate event data or a poor representation of the events. This is the closest precedent for the logic followed here: raw event records may add little, while a representation shaped by physiology may add more. Our absorption kernels are a light, non-dynamic analogue, and we claim no quantitative transfer from type 1 diabetes with insulin to free-living adults.

CGMacros [8] combines two simultaneously worn CGMs, activity-tracker data and photo-logged meals in 45 adults. Its own task is the two-hour response to a breakfast with leave-one-participant-out validation, following Zeevi et al. [23]; it reports no glucose-forecasting baseline. Our target, a continuously updated glucose value 30 minutes ahead, is a different task. The two devices differ in mean level (Section 3), and an inpatient study found only moderate agreement between the meal rankings of two simultaneously worn CGMs [12], so conclusions about meal-related factors may depend on the device. On CGMacros, a preprint adds a residual correction based on food images, macronutrients and meal timing to a frozen time-series foundation model and reports lower overall and post-meal errors (about 3% and 15% in the abstract) with a 24-hour context, a 2-hour horizon and a chronological within-subject split only [24]; its base model, horizon and metric differ from ours, so the numbers are not comparable.

The evidence that exogenous inputs improve forecasts beyond a strong glucose-history model is thus mixed and data-set dependent, which motivates adding factors to a fixed, adapted baseline and reporting paired changes under both protocols.

2.2 Automated feature engineering and screening

General-purpose systems automate feature construction for tabular learners; autofeat, for example, generates and selects nonlinear features for linear models [11]. OpenFE [25] is the most relevant here because its scoring rule matches the question asked in this study. It expands base features with unary and binary operators, including group-by statistics and frequency counts, and scores a candidate by the loss reduction obtained when a model fitted on the candidate is added to the predictions of a model trained on the base features (FeatureBoost). The group and frequency operators use other rows, which the paper analyses in a transductive setting; its Diabetes example is led by the number of admissions of the same patient. On overlapping CGM windows such operators would carry information across subjects and time, and in the released implementation the transformation step computes the new features on the concatenated training and test tables. We therefore use only row-wise operators and score candidates by their increment over the fixed baseline (Section 4.3).

tsfresh and its algorithm FRESH [6, 5] generate and then screen: a library of time-series characteristics is computed per series and each characteristic is tested separately against the target, with false-discovery control across the tests. Unlike our approach, FRESH screens features through univariate association tests rather than their incremental predictive value over a fixed baseline. We keep the generate-then-screen structure but screen candidates using participant-level summaries of their association with the baseline residual.

Formulaic factor mining in quantitative finance composes candidates from operators on price series [13]. AlphaForge [20] trains a generator against a surrogate of factor fitness (the information coefficient with future returns) and admits a formula only if it is weakly correlated with the factors already stored. T and L likewise use a finite operator grammar and a correlation filter, but are scored on training data against a fixed glucose baseline and use no neural generation.

2.3 LLM-assisted feature engineering

CAAFE [10] lets an LLM write Python code for new features from a description of the data set, with an explanation for each feature; the code runs under a syntax whitelist, and a feature block is kept if validation performance improves. LLM-FE [1] treats feature engineering as program search: the model proposes feature programs, each is scored by the downstream validation score, and programs are kept in islands and shown as in-context examples. Its prompts contain serialized data samples, and the authors list privacy of the samples and memorization of public data among the limits.

OCTree [16] asks an LLM for one new column at a time, as a natural-language rule translated into code; each new rule is proposed from the earlier rules, their validation scores and the text of a shallow decision tree fitted to the training data, and the rule with the best validation score after a fixed number of iterations is kept. ELLM-FT [9] writes a transformation as a postfix sequence over feature identifiers and a predefined operator set: reinforcement-learning agents fill a multi-population database, an LLM is prompted with performance-ranked sequences from one population, valid new sequences are scored with a fixed downstream model, and weak sequences and populations are culled. All four methods score candidates by the performance of a downstream model.

The LLM-guided search used here (L) also proposes candidates conditioned on scored earlier ones, but differs from these methods in its constraints. Proposals are expressions in a finite operator grammar over inputs observable at the forecast time, not free code, so every candidate can be audited against the availability rules; fitness is the gain over the fixed baseline on held-out parts of the training data, so validation and test data do not enter the search; the prompts contain no data rows; and the model proposes candidates only, with the final set chosen by forward selection on validation data (Section 4.4). Because the grammar is finite, a random-search control (Random-DSL) with the same number of unique valid evaluations separates what the model contributes from what the grammar and evaluator already provide. The restrictions limit expressiveness, and the evidence is a validation-data comparison with one model and one budget (Section 7.6).

2.4 Positioning of the present study

Table 1 compares the feature-construction methods reviewed above with the present study. The forecasting studies of Section 2.1 supply the task and the baselines but no factor search; the feature-construction methods supply candidate generation and scoring, but the general-purpose ones were largely evaluated outside the longitudinal, participant-grouped forecasting setting considered here. Apart from OpenFE in a restricted form, the published algorithms were not run; T and L are our own constructions that borrow ideas from them, so the contrasts are of design, not of results. None of the works read combines availability rules for lifestyle events, unseen-subject evaluation, the increment over a fixed adapted baseline and a comparison of manual, rule-based, automatic and LLM-guided candidates under one protocol.

3 Data, Task and Availability

3.1 The CGMacros data

CGMacros records approximately ten days from 45 adults: 15 without diabetes (HbA1c below 5.7%), 16 with pre-diabetes and 14 with type 2 diabetes (HbA1c above 6.4%) [8]. Participants with type 2 diabetes were excluded if treated with medicines other than metformin or with injectable GLP-1 agonists or insulin (original paper, p. 2). Two blinded CGMs were worn, an Abbott FreeStyle Libre Pro (native sampling every 15 minutes, upper arm) and a Dexcom G6 Pro (every 5 minutes, abdomen); a Fitbit supplied heart rate, activity calories and MET, and meals were logged with photographs and a food-logging application. Breakfast and lunch followed study-designed menus (a protein-shake breakfast and a restaurant lunch), at least three hours apart with only water or unsweetened coffee in between; dinners were freely chosen (pp. 2–3). Laboratory values, demographics and anthropometrics were collected at screening on the first day, and a subset of participants also provided a stool sample for gut-microbiome composition and gut-health scores. Libre is the primary history signal and target because it is more complete in the released rows (99.97% versus 91.6% for Dexcom). The authors report mean glucose of Libre 84.89, 105.2 and 138.04 mg/dL and Dexcom 122.36, 135.1 and 165.7 mg/dL in healthy, pre-diabetic and type 2 diabetic participants, with

Table 1: Feature-construction and screening methods reviewed in this section, compared with the present study. Entries restate what the cited papers report or are designed for.

Method	Candidate generation	Selection signal	Data setting
OpenFE	unary and binary operators, including group-by and frequency features	FeatureBoost: loss reduction over base-model predictions; two-stage pruning	static tables, independent rows
tsfresh / FRESH	library of time-series characteristics computed per series	univariate association tests with false-discovery control	time series, series-level samples
CAAFE	LLM writes feature code with explanations	accepted if validation performance improves	classification tables, random splits
LLM-FE	LLM proposes feature programs; island memory	downstream validation score fed back to the model	classification and regression tables
OCTree	LLM proposes a natural-language rule for one column	validation score and decision-tree summary fed back	general tables, two chronological regression tasks
ELLM-FT	LLM extends operator sequences drawn from RL-seeded populations	downstream-model score; culling of weak sequences	general classification and regression tables
AlphaForge	generator trained against a fitness surrogate	information coefficient with future returns; correlation filter	stock market data
This study	manual set (M); finite operator grammar (T) and LLM proposals within it (L); row-wise OpenFE (O)	training-data screening on the baseline residual (T), OpenFE score (O) or held-out gain (L); forward selection on validation data	CGMacros; within-subject and unseen-subject protocols (L and R: validation only)

Dexcom readings up to 58.7 mg/dL higher in healthy adults (p. 5; author-reported). The Libre target used here therefore has a low absolute level, with a hard floor at 40 mg/dL (Section 9), and conclusions about meal-related factors may depend on the device (Section 2.1).

3.2 Prediction task and windows

At each origin t, the task predicts Libre glucose at $t+30$ minutes using the closed history interval $[t-120, t]$. Origins are spaced five minutes apart, anchored to each participant's first released timestamp. A window requires non-missing released glucose throughout its history and at its exact target time, with no gap in released timestamps anywhere in $[t-120, t+30]$. These fixed rules admit 134,539 of 141,263 candidate windows (95.2%) before protocol-specific splitting; they exclude incomplete record boundaries, missing values and gap-crossing windows. The task is evaluated in two protocols, within-subject prediction of later periods (A) and prediction for unseen participants (B), which are defined in Section 6.1.

The original paper poses a different task: it predicts the two-hour area under the curve (AUC) and incremental AUC (iAUC) of a breakfast from anthropometric, biochemical, macronutrient and meal-start glucose predictors, leaving out one participant at a time [8]. Here the target is a continuously updated future glucose value, and additional factors are assessed relative to the project's glucose-history reference. Diet and core participant variables were already among the paper's PPGR predictors; activity was available in the data set but absent from that predictor set. A partial reconstruction of the PPGR task is given in Appendix A.3.

3.3 Alignment and prediction-time availability

The CGMacros authors linearly interpolated the Libre device's native 15-minute readings and the Dexcom device's native 5-minute readings onto a common one-minute grid before release (p. 4) [8]. Because the release does not identify the native sampling times, we infer candidate native anchors from changes in the slope of the interpolated series. Collinear readings may leave no detectable change, so the inferred anchors are not necessarily complete; Section 4.2 describes the rule and how uncertain states are handled. We preserve the released glucose values, align each participant's record to a one-minute index (inserting missing rows for absent timestamps), and perform no further interpolation or forward filling.

Interpolation can put later information into the glucose history: a value between two native readings depends on the later reading, which may occur after the forecast origin t. Under the inferred native-anchor schedule, the mean look-ahead in the training scope was approximately five minutes. Section 4 defines the native-glucose state used for new factors, and E8 (Section 7.7) examines the effect of restricting the baseline

history to inferred native readings.

Inputs are restricted to information at or before t, under explicit availability assumptions, because an event timestamp alone does not establish when a device reading or a log arrived.

- **Meals.** Meal times are taken from the released meal-log rows, which appear to correspond to the pre-meal photographs described in the original study (pp. 3–4). The recorded timestamps do not establish when the associated nutritional information became available. Breakfasts and lunches followed predefined meal designs, so their recorded nutrient values are treated as available at the logged time; this is an availability assumption rather than a verified property of the released records. Photograph-derived features and the consumed-proportion field are not used. Nutrient records for self-chosen dinners and snacks have unknown entry times. E9 examines this uncertainty by delaying when logged meal information becomes available by 30 minutes while retaining the original meal-event timestamps, for dinners and snacks only or for all meals. It is an availability sensitivity analysis, not a change to event timing or a verification of actual entry times. Modeling uses unmerged meal events with normalized meal types.

- **Activity.** The released MET values are documented as minute-level Fitbit estimates. The original paper mentions a 20-minute mean filter in its discussion of Figure 1 (p. 3) but does not establish whether that filter was applied to the released data. The manual set M uses MET windows ending at t, whereas OpenFE uses MET inputs lagged by at least ten minutes as a precaution. The theory-constrained and LLM-guided candidate generators (T, L and Random-DSL) use heart rate and activity calories instead.

- **Static data.** Static variables are joined by participant identifier. Laboratory values, gut-health scores and microbiome data were collected at screening and their return times are unknown, so they enter only offline extensions.

3.4 Manual feature groups and factor roles

Candidate factors were drawn from training-scope exploration and the data audit, then screened for availability at prediction time, coverage and data-quality ambiguity before any ablation. The role assigned to each modality, including the handling of HbA1c units, laboratory timing and photograph timestamps, is listed in Table 17 of Appendix D.1. Four manual feature groups are defined in Appendix D.1: G, ten glucose-history features (lags and trailing-window statistics); A, activity features with explicit coverage masks (missing activity is never coded as rest); D, diet features computed from the logged meals (nutrient totals are missing, not zero, when no meal is logged); and B, core participant variables (age, self-reported gender, BMI). The 47 activity and diet features of the primary variant together with the three core participant features (50 features in all) form the manual set M, the comparator in the factor-discovery experiments. HbA1c is excluded from the forecasting configurations of the formal comparisons; apart from the PPGR reconstruction (Appendix A.3) it enters only the follow-up analysis of laboratory-state interactions (E11).

4 Factor Discovery Framework

4.1 Aims, definitions and criteria

Factor discovery asks which additional inputs improve the forecasts of a fixed, adapted baseline, the GlucoBench linear regression (Section 5.2). A *candidate* factor is any generated feature; a *selected* factor belongs to the set chosen on validation data; and a *stable* factor is chosen frequently when the selection is repeated on subsamples of the training participants (E4), whether or not it belongs to the final selected set. These terms describe predictive selection and supporting evidence, not causal mechanisms (Table 18 of Appendix D.1 lists what each concept can and cannot support).

The manual set M is different in kind: its 50 features were specified in advance from domain knowledge (Section 3.4) and enter as one fixed block, with no screening or selection; it is the comparator. Only the factors of T and O (and of L and Random-DSL, validation only) are screened on training data and selected on validation data. A candidate counts as a *discovered* factor if it satisfies the availability rules of

Section 4.2 and enters the validation-selected block, whose formulas, fits and penalty are then frozen; the term is operational and does not imply a causal finding. For the reported formal comparisons, candidate screening and selection used training and validation data, after which the frozen blocks were evaluated on the test split for their increment over the fixed baseline, separately for A and B (Table 5; all 50 comparisons in Appendix H.1). Earlier uses of the test split are described in Section 6.4. Stability across subject subsamples and interpretability are supporting evidence, not thresholds.

Design rationale. Holding an adapted published forecaster fixed makes a gain mean new information relative to that forecaster rather than a better re-expression of glucose history; the fixed-baseline form is also how OpenFE passes baseline predictions as initial scores. Screening at the subject level with a fixed budget avoids window-level p-values that overstate significance when windows overlap, and follows the generate-then-screen pattern of time-series feature libraries [6] with the information-coefficient vocabulary of formulaic factor mining [13]. Subsampling-based selection frequencies in the spirit of stability selection [14, 19] are computed (E4) as descriptive measures only, because their error bounds assume conditions that repeated measures do not meet. Held-out improvement as the criterion follows the definition of importance as lost predictiveness [22], without claiming that the project's contrasts estimate that quantity.

4.2 Search space and availability rules

For the current operator whitelist, primitive inputs are restricted to observations at or before the forecast origin t. Source-offset bounds are propagated recursively through nested candidate expressions, and the validator rejects candidates whose latest source offset is positive. These bounds enforce the declared temporal semantics of the current whitelist; they do not establish when recorded data actually become available in practice (see Section 3.3). The legal inputs are:

- **Glucose state from inferred native readings only.** Native Libre readings are inferred from the released series as described in Section 3.3, using the fixed rule specified in Appendix D.2. The phase of the 15-minute sampling schedule at t is estimated using only slope changes (kinks) at or before $t - 15$ minutes (a kink at minute m depends on the reading at minute $m+1$); features at t use only values from inferred anchors at or before t. When the phase cannot be estimated reliably, the native-reading features are missing and a missing indicator is set; interpolated values are not substituted. The base model's 24 interpolated lags remain unchanged and are examined separately in E8.
- **Activity:** heart rate and activity calories (with coverage masks); MET only with a lag of at least ten minutes.
- **Meal events:** carbohydrate, protein, fat, fiber and calories as pulses at the logged meal time, by meal type (breakfast shake, lunch, dinner, snack); E9 also tests a version whose information availability is delayed by 30 minutes, without changing the logged timestamps.
- **Baseline state:** BMI (all participants) and, through the native-glucose state, the glucose level and slope before the meal. A family of 12 meal-by-laboratory-state candidates (fasting glucose, HbA1c and HOMA-IR across four meal types) was catalogued but not generated for the formal comparisons; laboratory values were reserved for offline analyses, and the family was evaluated afterwards on validation data only (E11, Section 7.8.2).

4.3 Candidate generation

M (manual comparator). The existing 50 features (Section 3.4), added as one block.

T (theory-constrained grammar). A fixed grammar of 206 candidates in five families, sized before any search and based on the predictors reported in the original paper (carbohydrate positive, protein and fat negative, fasting glucose and HbA1c among the main predictors; pp. 5–7), on meal-absorption models [3, 7], and on formulaic factor mining [13]. The families are *meal absorption* (the pulses of one nutrient of one meal type convolved with a gamma kernel of unit sum over ages 0–120 minutes), *meal-by-state interactions*

of the kernel-weighted carbohydrate with BMI or the native glucose state, *nutrient ratios*, *activity kernels* (summaries of heart rate and activity calories, including decaying and meal-kernel-weighted versions) and the *native glucose state*. Table 19 of Appendix D.2 lists the operators, grids and candidate counts.

O (OpenFE). The official OpenFE library [25] (version 0.0.12, library defaults apart from the initial scores) is run on the training windows with the baseline's out-of-fold predictions as the initial scores, on the 50 features of M except its eight unlagged MET features, together with the non-glucose primitives (including MET lagged by ten minutes), the baseline covariates and the native-state features, without the interpolated lags and without a subject column. Its row-wise operators (absolute value, logarithm, square root, square, sigmoid, rounding, residual, minimum, maximum and the four arithmetic operations) are kept; operators that use statistics of other rows (group-by, combine, frequency) are excluded because they would use information from other windows. The ranked formulas are then evaluated by project code. Because only row-wise operators are kept, no formula requires a fitted cross-row statistic, so O is a restricted OpenFE comparator.

4.4 LLM-guided search with a random control

L is an LLM-guided search for candidate factors; it generates candidates only, and the final factor set is chosen by the same validation procedure as for T and O (Section 4.5). Candidates were proposed by Claude Sonnet 5. Sampling parameters were not explicitly set in the recorded calls, and repeated calls are not guaranteed to return identical proposals; the archived requests and replies support replay and auditing of the recorded search (Appendix E). The model writes expressions, not free code, from the frozen operators of T over the same input signals, and a validator rejects any expression that reads records after the forecast time t; the limits of the records themselves (Section 3.3) apply to every candidate. Prompts include the task, the operator list and feedback from earlier proposals, but no data rows.

The search uses three branches with separate top-five prompt memories and a shared duplicate registry. Each branch runs for ten rounds with at most three proposals per call, giving 30 calls and at most 90 proposals per protocol; each prompt includes up to two previously accepted expressions sampled from the branch's memory, with their training gains. The fitness of an expression is its held-out gain in subject-equal RMSE over the fixed baseline, computed within the training split (Protocol A: the first 75% of each training period fits and the rest is held out; Protocol B: three-fold cross-validation grouped by participant over the 27 training participants). Validation data are used only afterwards, in the forward selection that chooses the final set, and test data are not used by either search.

The control, Random-DSL, draws from the same grammar and validator (one third single operators, two thirds products or ratios of two operators) and stops after the same number of unique valid fitness evaluations as the model search (67 in A and 70 in B); the numbers of calls and of proposals differ, since it makes no calls. Both searches pass their 25 best expressions by training fitness through the same redundancy filter (absolute Spearman correlation above 0.9 with the 24 baseline lag features or a kept expression), forward selection on validation data and one-standard-error rule as T and O. Because the grammar is finite, a random draw is a valid alternative proposer, so the control separates any contribution of the model from that of the grammar and the evaluator. L and Random-DSL were evaluated on validation data only (Section 7.6); the relation to CAAFE, LLM-FE and OpenFE is discussed in Section 2.

4.5 Screening and selection

Screening uses training data only and the subject as the unit. For each candidate the Spearman correlation with the baseline's out-of-fold residual is computed per training subject on windows down-sampled to a 15-minute step (to reduce autocorrelation), and the mean over subjects and the share of subjects with the same sign are recorded. The five candidates with the largest absolute mean correlation are kept per family (25 for T), then candidates with $|\rho| > 0.9$ against a baseline feature or an already kept factor are dropped (a pool of 15 for T in each protocol). p-values and false-discovery corrections of these correlations are reported only descriptively and never decide inclusion, because windows overlap and candidates are strongly correlated. Selection uses validation data only: forward selection adds at each step the factor with the largest decrease in validation subject-equal RMSE, up to ten factors, and the smallest set within one standard error of the

best validation curve is taken. This gave T_k with 6 factors in A and 7 in B and O_k with 7 and 4 (Table 20). The ridge penalty of every factor block is chosen on validation data on the grid $10^{-3}, \ldots, 10^{3}$ (13 values).

5 Models and Residual Augmentation

5.1 Baselines and comparison models

We implement two published forecasting methods for the CGMacros task, adapting their inputs to the released data (Appendix B.3). GlucoBench linear regression, the fixed reference for factor discovery, is global OLS with an intercept and 24 glucose lags spaced five minutes apart [18]. Sparacino AR(1)-WLS fits a local, intercept-free AR(1) model to glucose levels by exponentially weighted least squares and iterates it to the forecast horizon [21]; it uses three-minute samples of the released one-minute Libre series within the 120-minute task window.

Persistence repeats glucose at t, and linear trend adds 30 times the trailing 30-minute slope to that value; both serve as simple references. Ridge supports the ablation of G, A, D, B under a common fitting procedure, and shallow GBT checks the model family for G-only and full configurations. Their inputs and tuning are specified in Appendix B. Exploratory GRU/TCN checks are described in Appendix A.2.2, and the transfer of the frozen factor blocks to retrained GRU and TCN forecasters in Section 6.5.

5.2 Fixed-baseline residual augmentation

Let $\hat{y}_{\text{base}}(x_g)$ be the GlucoBench linear regression (24 lags), fitted on the training windows of a protocol and then frozen. For a factor block x_f (M, T, O or a combination), the residual $r = y - \hat{y}_{\text{base}}$ is computed on the training windows and a ridge model $f(x_f)$ is fitted on r; the prediction is $\hat{y} = \hat{y}_{\text{base}} + f(x_f)$. Imputation (median) and standardization of the factor block are fitted on training rows only. For a missing native-state estimate the block carries a missing indicator rather than an interpolated value.

Unlike a joint ARX-style fit, residual augmentation keeps the baseline's glucose-lag coefficients fixed. Joint fitting would allow these coefficients to change and could alter the estimated contribution of the additional factors. The residual form was chosen so that the baseline stays the published method and the increment is what the baseline cannot explain; the joint fit was not run. This residual formulation also matches the use of baseline predictions as initial scores in OpenFE.

Training residuals for screening, OpenFE and factor-block fitting are out-of-fold: temporal blocks within each participant in A and participant-grouped folds in B (Appendix B.4). Validation and test windows use the base fitted on the full training split. Factor-block penalties are chosen on validation data as in Section 4.5; boundary choices remain a limitation (Section 9).

6 Experimental Design

6.1 Protocols and cohorts

A: within-subject future prediction. Each participant's elapsed time span is divided approximately 60/20/20 into training, validation and test. A window's complete closed support $[t-120, t+30]$ must lie in one segment; training includes its right boundary, while validation and test are left-open and right-closed. Thus A predicts later periods of known participants.

B: unseen-subject generalization. The 45 participants form mutually exclusive 27/9/9 groups (seed 42); a held-out participant supplies only admissible prediction-time history, not fitting data. The split is not stratified by metabolic status or glucose measurements: counts below 5.7%, at 5.7–6.4% and above 6.4% HbA1c are 9/9/9 in training, 3/4/2 in validation and 3/3/3 in test.

Admitted training/validation/test windows number 79,300/26,330/26,269 in A and 81,958/24,259/28,322 in B (Table 38).

The primary *core* cohort requires G, A, D, B, including at least one heart-rate or activity-calorie reading in the history; *full* includes every admitted window with explicit missing-activity indicators and is supplementary, outside the multiplicity family. Core test counts are 44 participants/21,083 windows in A and 9/26,861 in B; full counts are 45/26,269 and 9/28,322. One A participant has no core window; 5,186 windows

(19.7%) are excluded. Evaluation subsets require at least 20 windows per participant (44 of 45 in A, all nine in B). Core and full scores cannot be subtracted to estimate feature gains.

6.2 Questions, hypotheses and comparison families

These questions and hypotheses were specified in the project plan before the factor experiments were run (Table 2). They concern prediction, not causal effects.

Table 2: Research questions, hypotheses and refutation conditions. These are planned interpretation rules; an unevaluated condition is not a negative result.

Question	Prediction, evidence and refutation condition
RQ1/H1	**Question:** Do diet, activity and baseline-metabolic features improve A/B forecasts, and do automated factors beat M?
	H1: A small gain (absolute subject-equal ΔRMSE below 0.3 mg/dL), concentrated after meals; absorption kernels outperform window sums.
	Evidence: P1–P4, S1–S2, E1, E5 and meal strata.
	Condition: H1 is refuted if all automated sets have intervals that include zero in both protocols or none beats the manual set, or if the gain is not concentrated in post-meal windows.
RQ2/H2	**Question:** Do fasting glucose, HbA1c and HOMA-IR help more as meal-by-state interactions than as additive terms?
	H2: Interactions add value in B, especially after meals; additive terms give none or mainly act as identity in A.
	Condition: H2 is refuted if interaction factors are not selected on validation, give no gain in B, or do not beat additive forms.
	Evidence: The laboratory-interaction family was not part of the formal test comparisons (Section 4.2); it was evaluated afterwards on validation data only (E11, Section 7.8.2). This is exploratory evidence, not a test of the planned comparison.
RQ3/H3	**Question:** How do T, O and L compare in gain, parsimony and interpretability?
	H3 (directional T/O expectation; exploratory L/R comparison): T is at least as good as O, which lacks temporal and event operators; an L advantage over R is uncertain.
	Evidence: P2/P4/S3 (T/O); E3 (L/R, validation only, equal unique valid fitness-evaluation counts; Section 4.4).
	Condition: H3 is refuted if OpenFE clearly beats the constrained constructor, or if the language model does not beat random search.
Validity	Interpolation may mask diet's increment; E8 restricts glucose history to inferred native readings and asks whether the increment is larger.
	This is not a separate RQ. The prediction is refuted if the increment is unchanged or smaller.

For RQ3, T versus O is judged on the test comparisons P2, P4 and S3, and L versus R, an exploratory comparison, on validation data only (Section 8.1).

The 50 formal test contrasts (24 in A, 26 in B) comprise four primary comparisons (P1–P4: the baseline with M added, with T added, M with T added, and the baseline with O added), three descriptive secondary comparisons (S1–S3: the baseline with M and T, with M and O, and T versus O), and the analyses E1, E5, E8 and E9 of Table 3 (seven E1 contrasts; E5 removes T families, two in A and four in B; E8 restricts glucose history to inferred native readings; E9 delays availability of self-chosen or all meal information by 30 minutes while retaining the logged timestamps). Full specifications and results are in Appendix H.1. The other analyses of Table 3 use training and validation data only (Sections 7.6, 7.7 and 7.8, Appendix F). The 36-participant grouped check uses B's 27 training and nine validation participants in six folds of six; these development participants do not provide independent external validation.

Table 3: Experiment and analysis codes used in the results. Data stage indicates where each analysis was evaluated.

Code	Experiment / analysis	Data stage
E1	Manual feature groups as residual blocks	Test
E3	LLM-guided versus random search	Validation
E4	Selection stability under subsampling	Training
E5	Removal of one theory-constrained (T) family	Test
E6	Subject-identity proxy sensitivity	Validation
E7	Selected and stable factor inventory	Training and validation
E8	Native-glucose-history sensitivity	Test
E9	Meal-log timing sensitivity	Test
E10	Frozen factor blocks added to fixed GRU/TCN forecasters (follow-up)	Training and validation
E11	Laboratory-state interactions (follow-up)	Training and validation

6.3 Metrics and uncertainty

MAE and RMSE (mg/dL) are calculated per participant and averaged equally (*subject-equal*); pooled-window metrics are secondary. Equal weighting prevents long records from dominating but can make participants with few windows influential. Paired ΔRMSE is the mean participant difference (to minus from) on identical windows; negative favors the added block. The project practical-effect rule labels a contrast improved (I) if $\Delta\text{RMSE} \leq -0.1$ mg/dL and more than half improve, worse (W) if $\Delta\text{RMSE} \geq 0.1$ mg/dL and fewer than half improve, negligible (N) if $|\Delta\text{RMSE}| < 0.1$ mg/dL, and mixed (M) otherwise; a participant improves when their difference is negative. This is not a clinical threshold or significance decision.

The 95% percentile intervals resample participants with replacement, keeping their windows and paired predictions together (2,000 resamples, seed 42). They describe fixed-prediction uncertainty, not retraining. Exploratory approximate bootstrap p-values are $p = \min\{1, 2(\min(N_{\geq 0}, N_{\leq 0}) + 1)/2001\}$ (minimum 0.001); Holm applies only to P1–P4 separately within each protocol and is also approximate. With only nine test participants in B, nominal 5% error and family-wise control are unverified: synthetic checks gave roughly 9–19% zero-effect rejection with nine subjects, 5–9% with 44, and about 17% Holm family error with nine. Effect size, improving share, the 0.1 mg/dL rule and intervals guide interpretation; neither Holm below 0.05 nor an interval excluding zero is called confirmation or significance.

Pre-specified descriptive meal strata use information before the origin: P30, a meal in $[t-30, t]$; P31–120, the last meal in $[t-120, t-31]$; NoMeal120, none in the past 120 minutes.

6.4 Data isolation and test use

Models and learned preprocessing were fitted separately for Protocols A and B. Shared exploratory analyses were restricted to Protocol A's training periods for the 27 participants assigned to training in Protocol B. Candidate generation and screening used training data, while factor selection and regularization parameters were determined using validation data (Figure 1). The baseline, residual formulation, candidate families and sizes, screening and selection procedures, evaluation strata, multiplicity adjustment, and primary comparisons were specified before factor fitting. All 50 planned comparisons, selected formulas, fitted models, and regularization parameters were finalized before formal test evaluation. Models were not refitted on the combined training and validation sets, and no changes were made in response to the formal test results. All planned comparisons, including those showing no improvement or worse performance, are reported.

The formal factor comparisons were evaluated once after the experimental choices were finalized, whereas the LLM-guided and random-search comparisons were restricted to validation data. Additional test records were accessed during data preparation and verification but were not used for factor screening, selection, or hyperparameter tuning in the factor study. Complete test-set isolation therefore cannot be established.

6.5 Follow-up Validation Analyses

After the formal comparisons, two further analyses were run on training and validation data only (E10 and E11 in Table 3). They are exploratory, are not among the 50 formal test comparisons and add no test evidence; the hypotheses and refutation conditions of Table 2 were not redefined for them, and their

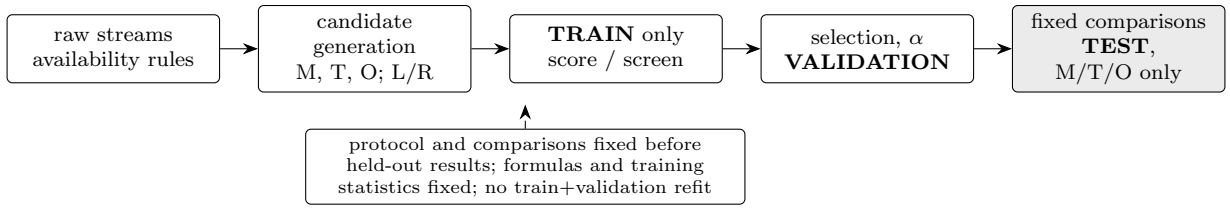


Figure 1: Protocol-specific data roles. Formal test comparisons use M/T/O; L/R and the follow-up analyses E10 and E11 stop at validation. Test features use fixed formulas and training statistics. Earlier test access is discussed in Section 6.4.

results only update the evidence status of H1–H3. Details and complete results are in Appendix F.2 and Appendix F.3.

Frozen-factor transfer to sequence baselines (E10). The compact GRU and TCN of Appendix A.2.2 were retrained as fixed baselines with five seeds each; the ensemble is the per-window mean of the five predictions, not the mean of five RMSEs. Training used a fixed number of epochs, identical for the out-of-fold (OOF) models and the full-training models, and no early stopping on the validation windows or on a held-out OOF fold (GRU 60 in A and 37 in B; TCN 35 in A and 34 in B). These numbers are the medians of the best epochs found by validation-based stopping in the earlier sequence checks, so the epochs depend indirectly on the validation split. OOF ensemble predictions on the training windows use temporal blocks within each participant in A and participant-grouped folds in B (Appendix B.4), with preprocessing re-estimated within each fold. In A, 5,117 of the 79,300 training windows have a 150-minute support that crosses a block edge; they have no OOF prediction and enter neither the OOF models nor the Ridge fit (74,183 legal windows), whereas the full-training sequence models, which produce the validation predictions, use all training windows. A Ridge model fitted on the OOF residual of the ensemble then adds the frozen blocks M, T_k and O_k without re-screening or reselection, so the factors correct the sequence forecast by residual augmentation (Section 5.2); no network is trained jointly with the factors. The Ridge penalty is chosen on the core validation windows (26,301 in A) on the grid of Section 4.5, and every model is reported on the same participants and windows (44 participants and 26,300 windows in A, nine and 23,139 in B; at least 20 windows per participant). The comparison of GlucoBench LR with the sequence models mixes architecture, input representation (25 glucose values against 24 lags) and training procedure, and is not a controlled architectural ablation.

Laboratory-state interactions (E11). Let R_{ref} be the fixed GlucoBench prediction together with the residual block of M, K_{meal} the four meal-kernel main effects (breakfast, lunch, dinner, snack), L_{lab} the three laboratory states (fasting glucose, HbA1c, HOMA-IR) and X_{int} their twelve products with the kernels. The main comparison is $R_{\text{ref}}+K_{\text{meal}}+L_{\text{lab}} \rightarrow R_{\text{ref}}+K_{\text{meal}}+L_{\text{lab}}+X_{\text{int}}$, so the interactions must add to the kernels and to the additive laboratory terms. What is fixed is the GlucoBench prediction, not the residual coefficients: every configuration refits its Ridge coefficients and selects its own penalty. In the primary construction (C-S1) the states are standardized with the mean and standard deviation of the training participants before multiplication; the sensitivity construction (C-S2) multiplies the unstandardized values. In A, a training identity proxy (each participant's mean OOF residual, one additional column) can be added to all configurations, as in E6. E11 differs from E6, which added four laboratory covariates (fasting glucose, fasting insulin, triglycerides, HDL) to M and is reproduced as a check (Appendix F.3). Laboratory results have unknown return times, so E11 is an offline analysis of predictive association, not a statement about prediction-time availability.

Evidence status. Intervals are subject-level paired percentile bootstrap intervals (2,000 resamples, seed 42) on fixed predictions; they exclude the uncertainty of training, epoch, penalty and factor selection and are not corrected for multiplicity. Test rows were excluded from all computations: mixed-source files that also contained test rows were read, and training and validation rows were isolated by their frozen keys before

any statistic, fit or selection. This isolation is statistical, not physical, and we do not claim that test rows were never decoded into memory.

7 Results

Unless stated otherwise, ΔRMSE is in mg/dL (to minus from; negative = lower error) on the core cohort with 95% subject-level bootstrap intervals (Section 6.3), and the test split is a previously used one.

7.1 Baseline Performance and Generalization

Table 4 compares two adapted published baselines and four reference/comparison models on all admitted test windows: 45 participants and 26,269 windows in A, nine participants and 28,322 in B. GlucoBench LR has the lowest subject-equal RMSE among these six models (10.83 mg/dL in A, 11.22 in B), with Ridge close behind; both outperform persistence. Shallow GBT lies between these linear models and persistence, whereas Sparacino AR(1)-WLS and linear trend fail to beat persistence. A forecasts later periods of known participants and B forecasts unseen participants; their scores do not compare identical windows. The restricted-core GlucoBench references, 10.957/11.327 mg/dL (A/B), are not interchangeable with the all-window scores.

Table 4: Test-set performance (mg/dL) for the adapted published baselines, simple references, and project comparison models, evaluated on all admitted test windows.

	Subject-equal		Pooled-window	
Model	MAE	RMSE	MAE	RMSE
Protocol A: 45 participants, 26,269 test windows				
Adapted published baseline				
GlucoBench linear regression	7.39	10.83	7.32	10.92
Sparacino AR(1)-WLS	10.60	16.63	10.71	17.11
Simple reference				
Persistence	10.04	15.02	10.12	15.56
Linear trend (30 min)	13.04	19.26	13.22	19.83
Project comparison				
Ridge (glucose history)	7.69	11.19	7.63	11.30
Shallow GBT (glucose history)	9.05	13.06	9.07	13.36
Protocol B: 9 participants, 28,322 test windows				
Adapted published baseline				
GlucoBench linear regression	7.43	11.22	7.43	11.27
Sparacino AR(1)-WLS	10.55	17.93	10.56	18.10
Simple reference				
Persistence	9.92	15.27	9.89	15.51
Linear trend (30 min)	12.80	19.46	12.80	19.61
Project comparison				
Ridge (glucose history)	7.64	11.37	7.64	11.40
Shallow GBT (glucose history)	8.94	13.27	8.94	13.43

A historical, post hoc comparison on exactly matched core windows (44 participants/21,083 windows in A; nine/26,861 in B) gave a paired ΔRMSE of full-factor Ridge C04 ($G{+}A{+}D{+}B$) minus GlucoBench LR of +0.335 [0.173, 0.509] in A and −0.202 [−0.693, 0.154] in B (mg/dL, 95% CIs): A favors GlucoBench LR, and B's interval spans zero. This comparison was outside the pre-specified formal family, and joint-Ridge comparisons are not interchangeable with fixed-baseline residual increments.

Exploratory sequence models. Compact GRU and TCN models trained on the same glucose history (five seeds each) reached RMSEs of 10.44 and 10.45 mg/dL in Protocol A and 10.73 and 10.72 in Protocol B.

Their paired differences from GlucoBench LR were −0.40 [−0.61, −0.21] and −0.38 [−0.62, −0.17] mg/dL in A, with intervals that excluded zero, and −0.49 and −0.51 in B, with intervals that included zero (Appendix A.2.2). These are exploratory comparisons outside the pre-specified evaluations: neither architecture had a supported advantage over the other, the networks were not part of the six main comparison models, and the formal factor comparisons were not run on them; the transfer of the frozen blocks to retrained networks is a separate validation-only analysis (Section 7.8.1).

Shallow GBT underperformed Ridge in all four matched configuration/protocol checks, with intervals excluding zero, under the tested tuning budget (Appendix C.2); the Ridge ablation of the manual groups is summarized in Section 7.3.

Post hoc error strata showed greater errors at high glucose and during rapid changes, with Diet-associated reductions concentrated in the first 30 minutes after a logged meal; this localization does not establish a meal mechanism (Appendix C.3). The released Libre readings have a floor of 40 mg/dL, so errors at this floor can be assessed only against the recorded target. Separately, in Protocol B most of the largest absolute errors of the full Ridge model came from one participant and coincided with large Libre–Dexcom discrepancies (details in Appendix C.3).

7.2 Primary Comparisons of Manual, Theory-Constrained and OpenFE Factors

Table 5 gives the eight pre-specified primary comparisons and the secondary comparisons; Figure 5 in Appendix G provides a visual summary of these contrasts. Five of the eight primary comparisons are smaller than the 0.1 mg/dL practical threshold (M, T and T added to M in A; T and T added to M in B); only OpenFE in A and the manual set and OpenFE in B reach the improved class. A Holm value below 0.05 cannot be read as confirmation (Section 6.3): these are pre-specified comparisons on a previously used test split, with approximate p-values, and Protocol B has nine test participants.

Table 5: Primary (P1–P4) and secondary (S1–S3) test comparisons on the fixed GlucoBench base (core cohort, 20-window rule). RMSE is subject-equal (mg/dL), from → to; ΔRMSE = to − from; Impr. = share of participants with lower error. p and Holm (P1–P4 only) are approximate bootstrap values (0.001 is the lowest reachable value), reported as exploratory; Label = practical-effect class (Section 6.3).

ID	Comparison	n	RMSE	ΔRMSE	95% CI	Impr.	p	Holm	Label
Protocol A, primary (Holm family of four)									
P1	base → base+M	44	10.957 → 10.910	−0.047	[−0.170, +0.084]	52%	0.462	0.462	negligible
P2	base → base+T	44	10.957 → 10.895	−0.063	[−0.120, −0.010]	64%	0.024	0.072	negligible
P3	base+M → base+M+T	44	10.910 → 10.877	−0.033	[−0.073, +0.013]	64%	0.159	0.318	negligible
P4	base → base+O	44	10.957 → 10.822	−0.135	[−0.229, −0.042]	59%	0.005	0.020	improved
Protocol A, secondary descriptive (no multiplicity adjustment)									
S1	base → base+M+T	44	10.957 → 10.877	−0.080	[−0.222, +0.075]	57%	0.306	–	negligible
S2	base → base+M+O	44	10.957 → 10.835	−0.122	[−0.256, +0.031]	61%	0.113	–	improved
S3	base+T → base+O	44	10.895 → 10.822	−0.073	[−0.143, −0.0002]	61%	0.051	–	negligible
Protocol B, primary (Holm family of four)									
P1	base → base+M	9	11.327 → 11.081	−0.245	[−0.362, −0.137]	89%	0.001	0.004	improved
P2	base → base+T	9	11.327 → 11.252	−0.075	[−0.116, −0.021]	78%	0.015	0.030	negligible
P3	base+M → base+M+T	9	11.081 → 11.017	−0.064	[−0.114, −0.007]	78%	0.033	0.033	negligible
P4	base → base+O	9	11.327 → 11.082	−0.244	[−0.349, −0.145]	100%	0.001	0.004	improved
Protocol B, secondary descriptive (no multiplicity adjustment)									
S1	base → base+M+T	9	11.327 → 11.017	−0.309	[−0.420, −0.190]	89%	0.001	–	improved
S2	base → base+M+O	9	11.327 → 11.025	−0.301	[−0.427, −0.183]	100%	0.001	–	improved
S3	base+T → base+O	9	11.252 → 11.082	−0.169	[−0.253, −0.098]	100%	0.001	–	improved

Protocols A (44 participants) and B (9 participants). Three points qualify this classification. In A the improvement of OpenFE is about one eightieth of the baseline RMSE (−0.135); the manual set has a point estimate of −0.047 with an interval that includes zero, and M+O is labeled improved by its point estimate (−0.122) although its interval also includes zero. T versus O is negligible in A (−0.073) and favors O in B (−0.169). In B the improvements of M and O (Table 5) rest on nine participants and should not be read as stronger evidence than the A results (Section 6.3).

Core versus full cohort. The full cohort is supplementary (Table 33 of Appendix G gives every primary, secondary and E1 contrast on both cohorts). In A the two disagree for the manual set: -0.047 on the core cohort (44 participants) and $+0.326$ $[+0.132, +0.542]$ on the full cohort (45 participants; only 31% of participants improve). The reversal is confined to contrasts that add the activity block (E1-1: $+0.360$; E1-6: $+0.403$); P2 to P4 keep their sign on the full cohort (-0.055, -0.037, -0.124). The two cohorts differ by 5,186 test windows in A (19.7% of the A test windows, including all windows of one participant) that have no activity record in the 120-minute history; from the saved pooled metrics, M raises the pooled RMSE on these additional windows by about 2 mg/dL while T and O do not (an arithmetic consequence of saved summaries, not a new evaluation). A possible explanation is that M's activity features, including coverage and missingness indicators, were fitted only on windows with activity observations and generalize poorly to windows without them; this was not tested. Both results are reported and neither is preferred; the reversal is a limitation of M. In B core and full agree (-0.245 and -0.236).

7.3 Existing Manual Groups: Ridge Ablation and Fixed-Baseline Blocks (E1)

Joint Ridge ablation. The manual groups Activity, Diet and Bio were added to a Ridge model fitted jointly on the glucose features G (42 paired contrasts; Appendix C). Its reference, Ridge on G (core-cohort subject-equal RMSE 11.30 in A and 11.48 in B), is a different model from the fixed base of the factor experiments, and its coefficients are re-estimated with every added group, so these increments are not on the same footing as the residual-block increments below. On the activity-restricted core cohort the three Diet contrasts reduced ΔRMSE by 0.27 to 0.30 mg/dL in B and no core interval excluded zero in A; the Activity and Bio contrasts were classified negligible in both protocols (Appendix C). On all admitted windows both Activity contrasts *worsened* Protocol A, with intervals excluding zero, a first sign of the cohort sensitivity of Section 7.2.

Fixed baseline (E1). Adding the manual groups as residual blocks on the fixed GlucoBench base (Table 6) gives a smaller and sparser picture than the Ridge ablation. In A all seven contrasts are negligible with intervals that include zero; the two contrasts that add Activity are slightly positive (marginally higher error). In B the three contrasts that add Diet (E1-2, E1-4, E1-7) reduce RMSE by about 0.20 mg/dL (classified improved), while the others are negligible. The diet finding of the Ridge ablation therefore survives a strong baseline for unseen participants, with a smaller magnitude, and not for within-subject prediction. On the full cohort the two Activity contrasts reverse in A (Section 7.2).

Table 6: E1: manual groups as residual blocks on the fixed baseline; core cohort, test, ΔRMSE [95% CI] in mg/dL. Superscripts: practical-effect class (N negligible, I improved, W worse); not significance decisions.

ID	Comparison	Protocol A	Protocol B
E1-1	Base → Base+A	$+0.026$ $[-0.045, +0.126]^N$	-0.017 $[-0.048, +0.011]^N$
E1-2	Base+A → Base+A+D	-0.051 $[-0.140, +0.029]^N$	-0.200 $[-0.324, -0.086]^I$
E1-3	Base+A+D → Base+A+D+B	-0.022 $[-0.060, +0.016]^N$	-0.028 $[-0.064, -0.002]^N$
E1-4	Base → Base+D	-0.053 $[-0.144, +0.026]^N$	-0.204 $[-0.345, -0.079]^I$
E1-5	Base → Base+B	-0.018 $[-0.053, +0.017]^N$	-0.023 $[-0.051, +0.001]^N$
E1-6	Base+D+B → Base+A+D+B	$+0.025$ $[-0.049, +0.125]^N$	-0.012 $[-0.059, +0.032]^N$
E1-7	Base+A+B → Base+A+D+B	-0.052 $[-0.141, +0.029]^N$	-0.202 $[-0.331, -0.086]^I$

7.4 Post-Meal Strata

Post-meal strata (pre-specified). The gain concentrates in the post-meal window in Protocol B and not clearly in Protocol A (Table 7). In B the P30 stratum has a ΔRMSE of -1.528 for M and -1.221 for O, against much smaller changes in the NoMeal120 stratum. In A the P30 point estimates are -0.199

(M) and −0.506 (O) but every P30 interval includes zero ($n = 25$ participants), and for T on top of M it is positive. The localization of the diet effect to the 30 minutes after a logged meal, first seen in the post hoc error analysis of the Ridge ablation (Appendix C.3), is therefore supported for unseen participants and undetermined for within-subject prediction, consistent with the post-meal part of H1 in B only.

Table 7: Post-meal strata of the primary comparisons (test, core cohort, 20-window rule; descriptive). n = participants with at least 20 windows in the stratum. P31–120 and NoMeal120 columns give ΔRMSE only, without intervals.

			P30	P31–120	NoMeal120
ID	Comparison	n	ΔRMSE [95% CI]	ΔRMSE	ΔRMSE
A:P1	base → base+M	25	−0.199 [−1.272, +0.977]	−0.006	−0.007
A:P2	base → base+T	25	−0.131 [−0.456, +0.211]	+0.056	−0.065
A:P3	base+M → base+M+T	25	+0.171 [−0.090, +0.550]	−0.092	−0.042
A:P4	base → base+O	25	−0.506 [−1.367, +0.422]	−0.179	−0.061
B:P1	base → base+M	9	−1.528 [−2.176, −0.849]	+0.044	−0.060
B:P2	base → base+T	9	−0.573 [−0.801, −0.321]	+0.097	−0.033
B:P3	base+M → base+M+T	9	−0.458 [−0.584, −0.329]	−0.001	−0.016
B:P4	base → base+O	9	−1.221 [−1.845, −0.519]	−0.132	−0.060

7.5 Selected and Stable Factors and Their Interpretation (E7, E4)

Nine candidate factors were selected in at least 60% of the half-samples of the stability analysis in at least one protocol; the complete list, with family, membership of T_k, the earliest source time of each formula and the leave-one-subject-out rate, is Table 34 (Appendix G). The selected sets of T_k and O_k and the formulas of O are in Appendix D.3.

The most stable factors are not absorption kernels: they are the native-glucose state (the last inferred native reading, `native_last`, chosen in 1.00 of the half-samples in A and 0.76 in B, with its slope over the last hour and its difference) and the activity statistics (the maximum and the mean of activity calories over 15 minutes), all stable in both protocols. Meal-by-state interactions and the carbohydrate share of the last meal are stable in one protocol only (Table 34). Only two members of T_k are stable in each protocol (2 of 6 in A, 2 of 7 in B), so a stable factor need not belong to the final selected set; the within-pair Jaccard index of the selected sets is about 0.3 in both protocols (Appendix F), and the absorption family is the least often represented in the selected sets: at least one absorption factor was chosen in 0.68 of the half-samples in A and 0.52 in B, fewer than for any other family (a family-level frequency, not the frequency of a single absorption factor). The discovered "factors" are therefore better read as a stable core of recent-glucose and recent-activity statistics plus an unstable set of meal terms. Stability describes how often a factor is chosen under subsampling; it does not make a factor causal.

Selection stability (E4). Selection stability was estimated with 50 complementary pairs of half-samples of the training participants; the numbers of stable factors at the three thresholds, the Jaccard indices and the overlap between A and B are in Appendix F (Figure 4). These frequencies are descriptive, not error rates, because repeated measures violate the assumptions of the stability-selection bounds (Section 4.1).

Figure 6 (Appendix G) shows, in training windows only, that the mean baseline residual rises with activity calories over 15 minutes and, with few windows and wide intervals, at the top deciles of the lunch carbohydrate by BMI interaction; these are descriptive in-sample associations, not mechanisms.

7.6 LLM-Guided Search versus Random Search (Validation Only)

The LLM-guided search (L) and its random control (Random-DSL, R) were run on training and validation data only and were not carried into the test evaluation (design: Section 4.4). Their numbers are validation-stage estimates, selected on the same split, and are optimistic; they are not test conclusions. Table 8 gives the stored records. The two searches are matched on the number of unique valid fitness evaluations (67 in A and 70 in B), not on calls, proposals or cost (L made 30 model calls per protocol, R none).

Table 8: LLM-guided search (L) and its random control (R): stored records. The two searches are matched on unique valid fitness evaluations, and Random-DSL makes no model calls. Training fitness is the held-out gain in subject-equal RMSE (mg/dL) on training data only; validation ΔRMSE (mg/dL) is optimistic because the factors were chosen on the same data. All selected factors: Tables 22 and 23 (Appendix E).

	Protocol A		Protocol B	
	L	R	L	R
Model calls	30	0	30	0
Proposals or random draws (all valid)	87	70	90	74
Duplicates (share of proposals)	20 (23.0%)	3 (4.3%)	20 (22.2%)	4 (5.4%)
Unique valid evaluations	67	67	70	70
with positive training fitness	54/67	33/67	2/70	5/70
Best training fitness	0.0425	0.0959	0.0474	0.0341
Mean training fitness of the pool of 25	0.0163	0.0113	−0.0056	−0.0025
Kept after the redundancy filter / selected on validation (k)	10 / **6**	18 / **4**	15 / **5**	19 / **8**
Validation ΔRMSE, base → +set [95% CI]	−0.113 [−0.175, −0.050]	−0.147 [−0.223, −0.069]	−0.155 [−0.276, −0.034]	−0.166 [−0.286, −0.045]
L relative to R	+0.034 [−0.013, +0.081]		+0.012 [−0.065, +0.075]	

Search behavior. The model's proposals contained many more duplicates than the random draws (Table 8), although the prompts listed recent rejections; this describes how the model's proposals concentrate, not whether they are better. All proposals were valid; one reply in A could not be parsed, which is one call without proposals and not three invalid factors. Training fitness differs in direction between the protocols and did not decide membership of the final sets: three of the five selected L factors and five of the eight selected R factors in B have a negative training fitness, because forward selection on validation data chose them.

Selected expressions and overlap with T and O. The selected L expressions include heart-rate summaries, carbohydrate load interacted with glucose momentum or divided by the current glucose level, meal timing, and carbohydrate per BMI, each with the hypothesis the model stated (Table 22, Appendix E); these are motivations, not findings, and the logs contain no test of them. For the 11 selected L expressions the nearest factor of T or O (absolute Spearman correlation) was a T factor for 6 and an O factor for 5, with a median of 0.60 (range 0.39 to 1.00); one expression is stored as a near-duplicate of a T factor (correlation reported as 1.00 at the displayed precision). Several expressions thus resemble families that T and O already contain; this is a nearest-neighbor statement and says nothing about overlap with M, for which no analysis exists.

Validation comparison. Added to the fixed baseline, L and R both reduce validation ΔRMSE in both protocols (Table 8); L relative to R is positive in both, i.e. the point estimates favor R, with intervals that include zero, and O was better than L in A (+0.073 [+0.022, +0.124]). Added on top of M instead, L is ahead of R in A with an interval that only just excludes zero, and not in B (Table 24, Appendix F); the differences between L and R and between O and L are all below the 0.1 mg/dL practical threshold. The LLM therefore showed no consistent advantage over a random search with the same number of evaluations on validation data. A random draw from the same grammar reached validation gains of the same size and both searches selected expressions from the same few families; the records cannot separate a small benefit of model guidance from none, because there is one search per protocol with one model, one prompt template and one fixed budget, and selection and evaluation share the validation data (Section 8.1).

7.7 Sensitivity and Robustness (E5, E6, E8, E9, Validation and CV)

Shrinkage from validation to test and the 36-participant CV. Table 9 compares the three evidence stages. In A the validation gains shrink on test, as expected when validation was also used for selection and

for the ridge penalty; this mixes winner's-curse shrinkage and drift and cannot be separated here. In B the test gain of M (−0.245) is larger than on validation (−0.164) and larger than in the 36-participant grouped CV (−0.071), so the B test value for M is probably on the high side of what the data support. The effect of T is consistent across the three stages (all below 0.1 in B). The CV uses six folds of six participants with screening and selection repeated inside each fold; it includes participants used in early development and is not an independent external validation.

Table 9: Validation, 36-participant grouped CV (Protocol B only) and test ΔRMSE [95% CI], mg/dL. Validation is the selection split; the CV and test intervals are bootstrap intervals.

ID	Comparison	VALIDATION	36-subject CV	TEST
A:P1	base → base+M	−0.153 [−0.265, −0.040]	–	−0.047 [−0.170, +0.084]
A:P2	base → base+T	−0.103 [−0.153, −0.054]	–	−0.063 [−0.120, −0.010]
A:P3	base+M → base+M+T	−0.098 [−0.131, −0.065]	–	−0.033 [−0.073, +0.013]
B:P1	base → base+M	−0.164 [−0.302, −0.018]	−0.071 [−0.162, +0.027]	−0.245 [−0.362, −0.137]
B:P2	base → base+T	−0.080 [−0.136, −0.027]	−0.061 [−0.092, −0.030]	−0.075 [−0.116, −0.021]
B:P3	base+M → base+M+T	−0.023 [−0.059, +0.014]	−0.056 [−0.079, −0.030]	−0.064 [−0.114, −0.007]

Which T families matter (E5). Removing one family from T changes the error by at most +0.044 in A (activity kernels; meal-by-state +0.012) and by +0.009 (absorption), +0.023 (nutrient ratios), −0.008 (activity) and +0.032 (meal-by-state) in B (Table 37, Appendix H.1); all are negligible. Among these tested family removals, the small gain of T is associated mainly with the activity statistics in A and with the meal-by-state and nutrient-ratio families in B; the native-glucose-state family was not separately ablated. The absorption and activity-kernel removals produce negligible changes in B, with intervals that include zero.

Timing of meal logs (E9). In A the manual set M deteriorates when the self-chosen meals (dinner, snack) are made available 30 minutes later: +0.434 [+0.227, +0.710], a worse class, and +0.061 when all meals are delayed; in B delaying all meals gives +0.192 [+0.089, +0.307] (worse class) and delaying only the self-chosen meals +0.003. T is insensitive (at most +0.069, below 0.1). In A, M is worse under the smaller delay than under the larger (non-monotone); the saved results do not identify why. M's gain in A therefore depends on the assumption that meal records are available at the logged time, and this assumption is not verified for self-chosen meals. The intervals are in Table 37 (Appendix H.1).

Table 10: E8: test ΔRMSE [95% CI], mg/dL, core cohort, reliable windows. RMSE levels, validation and full-cohort contrasts, and the E5 and E9 contrasts are in Table 37 (Appendix H.1).

ID	Comparison	Protocol A	Protocol B
E8: history from inferred native readings only (anchor-causal base); reliable windows, $n = 43$ in A and $n = 9$ in B			
E8-1	original base, reliable windows → anchor-causal base	+4.163 [+3.771, +4.584]	+4.109 [+3.587, +4.701]
E8-2	anchor-causal base → anchor-causal base+M	−0.280 [−0.498, −0.062]	−0.739 [−0.989, −0.497]
E8-3	anchor-causal base → anchor-causal base+T	−0.115 [−0.199, −0.033]	−0.256 [−0.373, −0.119]
E8-4	anchor-causal base → anchor-causal base+O	−0.299 [−0.500, −0.079]	−0.556 [−0.751, −0.400]

History restricted to inferred native readings (E8). The base is rebuilt with its 24 lags taken from the newest inferred native reading at or before each lag time (last reading held; at least nine anchors), refitted on the training windows, and compared on windows with a reliable anchor estimate (43 participants and 20,800 windows in A, 9 and 25,935 in B; 1.3% and 3.4% of core windows excluded). The restricted base is worse by +4.163 mg/dL in A and +4.109 in B, and every participant gets worse (Table 10; RMSE levels in Table 37 of Appendix H.1). On the restricted base, M, T and O give larger gains than on the original base (Table 10), but the restricted base with a block is still far worse than the original base (about 14.8–14.9 versus 10.9 in A). The prediction that gains are larger is therefore not refuted, but a weaker base leaves more

room for any block, so this is not evidence that diet factors are more useful. A matching check on validation data (the original lags refitted on the same reliable training windows) gave a change of about 0.001, so the deterioration comes from rebuilding the lags and not from the smaller fitting set. E8 cannot separate the loss of readings taken after t (interpolation look-ahead) from the loss of information density and the error of the inferred anchors; it is a sensitivity analysis, not a verification of the native sampling times or a measurement of leakage.

Subject-identity proxy (E6; validation only). Adding a subject-level mean-residual proxy in A removes the validation gains of the static and laboratory blocks (optimistic estimates, as in Section 7.6): core static features -0.013 without and $+0.0002$ with the proxy; laboratory values (fasting glucose, fasting insulin, triglycerides and HDL; a different set from the three states of E11) -0.078 and -0.0002; categorical gut scores -0.118 and -0.006. This is consistent with these "gains" being identity effects.

7.8 Follow-up Validation Analyses

The two analyses of this subsection (Section 6.5) were run after the formal comparisons, use validation data that were also used for development and selection, and are exploratory; they provide no test evidence and no new multiplicity-controlled inference.

7.8.1 Frozen-factor transfer to sequence baselines (E10)

On the shared core validation windows the fixed GRU and TCN ensembles had a lower subject-equal RMSE than GlucoBench LR in both protocols, and all four paired intervals lie below zero (Table 26 of Appendix F.2). This comparison mixes architecture, input representation and training procedure and is not a controlled architectural ablation, and its values are not comparable with the single-seed test RMSEs of Section 7.1, which refer to a different split and training procedure.

Table 11 shows what the frozen blocks add to these ensembles. All twelve point estimates are negative (-0.084 to -0.162 mg/dL) and eleven intervals lie below zero; the exception is M with the GRU in B (-0.125 [-0.270, 0.021]; nine participants). O has the largest point estimate in all four network–protocol settings, and the point estimate of T is below 0.1 mg/dL with the TCN in both protocols. Compared with the same blocks on the linear baseline (last column), the point estimates of M and O are smaller on the sequence ensembles in all four settings, whereas T shows no consistent change; none of these differences between networks, blocks or bases was tested. The blocks were not reselected for the sequence models, but T_k, O_k, the epochs and the penalties depend on the validation split used here, and O contains formula types beyond lifestyle variables. The result indicates that the small gains of Section 7.2 do not disappear on a stronger forecaster in these validation data; it is neither an independent confirmation nor a test result.

7.8.2 Offline laboratory-state interactions (E11)

In the primary construction without the identity proxy, adding the twelve interactions to a model that already contained the meal kernels and the three laboratory states changed validation RMSE by -0.019 [-0.052, 0.013] mg/dL in A (44 participants, 26,300 windows) and by -0.023 [-0.076, 0.030] in B (nine participants, 23,139 windows; Table 29 of Appendix F.3). Both intervals include zero and both point estimates are far below 0.1 mg/dL: the data show no clear overall gain from the interactions, which is not evidence that they are useless or equivalent to additive terms. The four other settings (C-S2; with the identity proxy in A) gave point estimates between -0.017 and -0.026 and intervals that also included zero. The proxy did not remove the small point estimate in A (-0.019 [-0.053, 0.012]), unlike the gain of additive laboratory values in E6, which disappeared with the proxy (Section 7.7); a single residual-mean column, however, does not exclude all identity effects.

In the first 30 minutes after a logged meal the point estimates were larger, -0.116 [-0.222, -0.010] in A and -0.467 [-0.946, 0.019] in B (eight participants), but the A interval includes zero with the proxy or with C-S2 (Table 31), the B interval includes zero, and the estimates 31–120 minutes after a meal are positive. An auxiliary screening and forward selection chose three dinner interactions (with HbA1c, HOMA-IR and fasting glucose) in A, with a small gain whose interval barely excludes zero, and none in B; selection with an unpaired one-standard-error rule chose none in any setting, and the selection and its evaluation use the

Table 11: E10: frozen factor blocks added to the fixed GRU and TCN ensembles (validation data, core cohort; A: 44 participants and 26,300 windows, B: nine participants and 23,139 windows). ΔRMSE [95% CI] in mg/dL (to − from; negative = lower error) relative to the sequence ensemble; Impr. = share of participants with lower error; the last column gives the same block added to GlucoBench LR on the same windows (point estimate). Exploratory validation estimates, optimistic; the intervals are those described in Section 6.5.

Network	Prot.	Block	ΔRMSE [95% CI]	Impr.	On LR
GRU	A	M	−0.114 [−0.215, −0.010]	57%	−0.153
GRU	A	T_k	−0.102 [−0.161, −0.045]	61%	−0.103
GRU	A	O_k	−0.162 [−0.260, −0.062]	64%	−0.186
GRU	B	M	−0.125 [−0.270, +0.021]	67%	−0.164
GRU	B	T_k	−0.101 [−0.169, −0.031]	78%	−0.080
GRU	B	O_k	−0.156 [−0.246, −0.073]	89%	−0.222
TCN	A	M	−0.108 [−0.203, −0.008]	55%	−0.153
TCN	A	T_k	−0.086 [−0.145, −0.032]	61%	−0.103
TCN	A	O_k	−0.130 [−0.219, −0.040]	59%	−0.186
TCN	B	M	−0.124 [−0.236, −0.015]	78%	−0.164
TCN	B	T_k	−0.084 [−0.148, −0.017]	78%	−0.080
TCN	B	O_k	−0.135 [−0.218, −0.054]	89%	−0.222

same validation data (Table 32). These signals depend on the construction, the proxy and the selection rule; they are exploratory predictive associations, not evidence of a physiological interaction.

8 Discussion

8.1 Answers to the Research Questions

The following findings are interpreted in relation to the pre-specified hypotheses rather than as claims of statistical significance.

RQ1/H1. The evidence is mixed across the components of H1:

- **Small gains.** Adding diet, activity and baseline information to the fixed baseline gives small gains; all sets except two combinations in B have an absolute ΔRMSE below 0.3 mg/dL (−0.309 and −0.301 slightly exceed it), and Diet gives the clearest gain, in B only. This part of H1 is supported.
- **Concentration after meals.** The gain is concentrated after meals in B, but the evidence is inconclusive in A. In A every P30 interval includes zero and no analysis compares the strata directly, so the data neither show nor exclude a concentration (Section 7.4).
- **Absorption kernels versus window sums.** The expected advantage of absorption kernels is not supported and was not directly tested. T is not clearly better than M (A: −0.063 versus −0.047, no paired comparison; B: −0.075 versus the larger −0.245), T on top of M is negligible, and removing the absorption family from T changes B by only +0.009 (it was not removed in A). T and M differ in far more than kernels versus window sums, so these results do not isolate the kernels and do not show that they are worse.
- **Automated versus manual factors.** The results do not support an advantage for automated factors in B and are mixed in A. In B the point estimates of O (−0.244) and T are not better than that of M (−0.245); in A those of T and O are better (−0.063 and −0.135 versus −0.047). M has no paired comparison with either.

The pre-specified refutation conditions require separate interpretation. "All automated sets have intervals that include zero in both protocols" is not met. "None is better than the manual set" is met by the point

estimates in B and not in A, without a paired comparison in either. "The gain is not concentrated in post-meal windows" is not met in B and is not established in A, where the evidence is insufficient rather than contrary. Read literally, the second condition counts against H1 in B; because it rests on unpaired point estimates, it is reported as non-support, not as a formal refutation. In the validation-only follow-up E10, the frozen M, T and O blocks also lowered the error of fixed GRU and TCN forecasters by small amounts (Section 7.8.1), so the small gains are not confined to the linear baseline; this is exploratory evidence, limited by the repeated use of the validation split, and does not change the assessment of H1. The gains depend on protocol, cohort and factor class, without demonstrated clinical meaning.

RQ2/H2. The central interaction-versus-additive comparison in H2 was not part of the formal test comparisons: the 12 meal-by-laboratory-state candidates (fasting glucose, HbA1c and HOMA-IR across four meal types) were catalogued but not generated under the frozen protocol, which kept laboratory values for offline analyses. They were evaluated afterwards in E11, on validation data only (Section 7.8.2): added to a model that already contained the meal kernels and the three additive laboratory states, the interactions gave no clear overall gain (small point estimates favoring the interactions, intervals including zero), so the follow-up gives no clear support for H2 under the evaluated settings. Read literally, two refutation conditions of Table 2 are met in B (no interaction is selected by the paired validation rule, and no clear gain beyond the additive forms), but the analysis is exploratory, reuses the validation data for selection and evaluation and does not establish equivalence, so it is reported as non-support, not as a refutation; no independent test of the interactions exists. The post-meal and selection signals in A depend on the construction, the identity proxy and the selection rule. Because static or laboratory features can act as subject identifiers in Protocol A but not in Protocol B, the same factor can mean different things in the two protocols; the original sensitivity analysis E6 (four laboratory covariates added to M) is consistent with an identity-related explanation for additive laboratory features (Section 7.7) but does not address interactions. The interactions evaluated in the formal comparisons used BMI or the native glucose level and slope rather than laboratory measurements; removing that family from T changes the error negligibly in both protocols (Section 7.7).

RQ3/H3. Two comparisons address different aspects of H3:

- **Theory-constrained construction versus OpenFE.** The expectation that T is at least as good as O is not supported: T versus O is negligible in A (the upper interval limit essentially at zero) and favors O in B (Section 7.2). The difference is between two configurations as run, not between the methods: OpenFE's inputs include the manual features, the two searches differ in space and procedure, and only B has a gap above the practical threshold (with nine participants).
- **LLM-guided versus random search.** The comparison is exploratory, uses validation data only (unlike the test-based T–O comparison) and shows no consistent advantage of L: added to the baseline, the point estimates favor R in both protocols, so the refutation condition "the language model does not beat random search" is met literally; added on top of M, L is slightly ahead of R in A only (Section 7.6). The model's role is to propose readable expressions under the fixed grammar and availability rules, whereas validation-based forward selection determines the final subsets; random proposals from the same grammar also gave validation gains, so the design cannot separate the contribution of model guidance from that of the grammar, the evaluator and the selection procedure (Section 9).

O's selected formulas are short arithmetic combinations, whereas T's formulas carry physical units and kernel meaning but are less stable under subsampling. In E10, O has the largest point estimate of the three blocks in all four network–protocol settings, but the intervals overlap, no paired comparison among the blocks was made, the validation split was reused and O contains formula types beyond lifestyle variables, so this does not establish that OpenFE is the better method.

Validity check. As predicted, the gains are larger on the anchor-causal base; because that base is far worse, the prediction is not refuted but the result is inconclusive and is not evidence for a larger information value of diet factors (Section 7.7).

8.2 Relation to the Literature

The small and inconsistent gains found here are consistent with the finding of GlucoBench that the value of covariates varies across data sets [18]. Unlike the benefit that Bunescu et al. [3] report for physiologically shaped features, T did not clearly outperform M, although our comparisons do not isolate absorption kernels and the study settings differ. The concentration of diet-related gains after meals in Protocol B is directionally consistent with results reported for a frozen time-series foundation model on CGMacros [24], despite differences in prediction task and evaluation. Restricted OpenFE produced the largest gain in Protocol A among the factor sets considered. A directional comparison with the original paper's SHAP findings is given in Table 35.

9 Limitations

Sample and statistics. There are 45 participants, nine in the unseen-subject test and 44 in the within-subject core test; overlapping windows add no independent participants, and subject-equal weighting lets participants with few windows weigh more. The p-values are approximate, Holm is not a verified family-wise control, the 50 comparisons were not corrected jointly, and the 0.1 mg/dL rule is a reporting convention, not a clinical threshold. The test split was previously used in project development, so the factor comparisons should be interpreted as descriptive rather than independently confirmatory.

Baselines and models. Sparacino AR(1)-WLS is an adaptation (three-minute samples from the author-interpolated one-minute grid, a different device and population), not a like-for-like replication. The ridge and boosting grids are fixed and several selected values lie at their edges ($\alpha = 1000$ for M, 0.001 for T and O), so no block is fully tuned, the shallow GBT result is limited to its matched tuning budget (two candidate settings) and does not rule out other nonlinear methods, and residual augmentation replaces a joint refit that might give larger gains. Factor increments were evaluated on a fixed linear baseline in the formal comparisons; the transfer to GRU and TCN forecasters (E10) is a validation-only follow-up whose epochs, frozen factor sets and penalties depend on the validation split, and the GRU/TCN test checks that outperformed that baseline within subjects are exploratory (Section 7.1); the C04 comparison is post hoc. Small RMSE differences have no demonstrated clinical meaning.

Measurement, timing and availability. Interpolated history can contain readings taken after t (mean look-ahead 5.00 minutes in the training scope), meal-log times may follow the start of eating (E9 shows that the manual set depends on this assumption), and complete-history admission leaves accuracy during sensor gaps untested; the availability rules of Section 3.3 therefore do not show real-time availability. E8 cannot separate the removed look-ahead from the loss of information density and the error of inferred anchors (Section 7.7); it is a methodological limit of the baseline, not a measurement of leakage. The Libre target is systematically lower than Dexcom (Section 3) and is floored at 40 mg/dL; device robustness was checked only for the GlucoBench baseline, on a Dexcom target (Appendix A.2.1), and not for the factor experiments.

Factor discovery and interpretation. Selection stability is not causal stability, and kernels, interactions and other expressions are statistical forms, not identified mechanisms; correlated features can substitute for one another, and static or laboratory features can act as subject identifiers in A. The sets T_k and O_k depend on the selection rule, T, O, L and R were searched in different spaces on non-identical inputs, and core-cohort conclusions for M do not hold on the full cohort in A. The laboratory-state interactions (RQ2) were examined only in the validation-only follow-up E11, an offline analysis because laboratory results have unknown return times, in which a single identity proxy cannot exclude all identity effects.

Follow-up analyses (E10, E11). They add no test evidence: the validation split had already been used repeatedly for development and for the selection of T_k, O_k, epochs and penalties, and E11 selected and evaluated its auxiliary subset on the same data. The bootstrap intervals condition on fixed predictions and exclude the uncertainty of training and selection, nominal 95% intervals are not adjusted for multiplicity,

and Protocol B has only nine validation participants (eight in some strata). The isolation of test rows was statistical, not physical (Section 6.5).

LLM results. L and R were compared on validation data only, with one model, one prompt template, one budget and one search per protocol, and with selection and evaluation on the same validation data; the contributions of the grammar, the evaluator and the selection procedure were not separated from that of model guidance (Section 8.1).

10 Future Work

Three directions follow from the findings of this study.

Independent evaluation. The factors that were selected or were stable under subsampling (recent native glucose state, short-window activity energy, meal-by-state terms), and the validation-only findings on sequence forecasters and laboratory-state interactions, should be tested on new participants or larger CGM data sets with diet records, with timestamps and availability recorded at collection and with Dexcom as a second target for the device dependence reported by Howard et al. [12].

Individualized and quasi-experimental factors. The designed breakfasts repeat the same recipes on several days (original paper, Table 3), so a participant's earlier response can be tested as a predictor of the later one (an interaction with the individual breakfast response gave no gain on validation data; Appendix F), and the designed-menu subset allows composition effects to be estimated with less confounding than from free choices.

Controlled automated search. An LLM-guided search should be compared with random search over several seeds, models and budgets, with model-assisted candidate or feature screening and a test evaluation fixed in advance.

Several questions also remain open: a strictly nested evaluation of the laboratory-state interactions with prediction-time laboratory availability (RQ2), device robustness, a causal-interpolation control that separates look-ahead from information density in E8, a per-participant decomposition of the full-cohort reversal in A, and a joint ARX comparison.

11 Conclusion

This study adapted established glucose-forecasting baselines and assessed whether manual and automated factors improve 30-minute prediction beyond glucose history. GlucoBench was the strongest of six main comparison models. Added factors yielded small, protocol-dependent gains, with diet-associated improvements most apparent after meals for unseen participants. Theory-constrained factors did not outperform OpenFE, and LLM-guided search showed no consistent advantage over random search on validation data. Laboratory-state interactions also showed no clear validation-only benefit. Because the test split had been used previously and the unseen-subject test comprised only nine participants, these findings remain descriptive and require independent validation.

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A Supplementary Gut, Device, Sequence-Model and PPGR Analyses

A.1 Gut-Health and Microbiome Analyses

Gut-health scores. Ordinal and categorical encodings of 22 gut-health scores were compared with the full configuration refitted on the same Gut-available cohort: 41 participants in A and 8 in B. Ordinal encoding changes test subject-mean RMSE by −0.14 mg/dL (95% CI −0.27 to −0.01) in A and +4.04 (1.41 to 6.66) in B; categorical encoding changes it by −0.22 (−0.37 to −0.07) and +1.25 (0.58 to 1.96), respectively. Both encodings improve within-subject prediction and worsen unseen-subject prediction with intervals excluding zero, specifically on this restricted cohort.

Microbiome extension. A subject-level Ridge correction of the full-model residuals with binary microbial detection indicators (screening and fitting on training participants, penalty chosen on validation data) was compared with the full model refitted on the matched microbiome–Gut test cohort of 41 participants/19,888 windows in A and 8/23,455 in B. Test ΔRMSE is −0.20 mg/dL in A and −0.01 in B (validation: −0.17 and +0.02). This small within-subject change does not demonstrate a transferable benefit for unseen participants.

A.2 Exploratory Device, Horizon, and Sequence-Model Checks

A.2.1 Device and Horizon Sensitivity

Two tasks retain the history duration and model specification but use Dexcom history/target at 30 minutes or Libre at 60 minutes. GlucoBench LR beats same-task persistence in all four task/protocol comparisons (Table 12). Relative to primary-task test-window counts, retained coverage is 61.6%/88.3% for Dexcom and 98.9%/99.6% for Libre at 60 minutes (A/B). Absolute errors are higher, but device, horizon, coverage, and sample composition prevent interpreting these differences as a controlled change in model accuracy.

Table 12: Exploratory test-set performance (subject-equal MAE/RMSE, mg/dL) for GlucoBench linear regression under two supplementary task definitions.

Task	Protocol	MAE	RMSE	ΔRMSE*
Dexcom target, 30 min	A	11.94	16.65	−1.34
Dexcom target, 30 min	B	11.74	17.49	−2.11
Libre target, 60 min	A	14.47	20.39	−2.50
Libre target, 60 min	B	14.43	20.93	−2.95

*Relative to a same-task persistence reference (subject-equal RMSE); negative values indicate lower error than persistence.

A.2.2 Compact Sequence Models

A compact GRU (3,393 parameters) and a causal TCN (2,465 parameters) were each trained with five fixed seeds (0–4) on 25 glucose values spaced five minutes apart across the history, predicting the difference between the target and the latest value. The GRU has one layer with 32 hidden units; the TCN has four causal convolution blocks (16 channels, kernel size 3, dilations 1, 2, 4 and 8). Both used Adam (learning rate 10^{-3}, batch size 512), at most 60 epochs and early stopping on validation loss (patience 6), and were evaluated on the same test windows as the formal baselines. Four GRU seeds in A reached the epoch limit; the budget was not extended. The GRU was trained on a GPU and the TCN on a CPU, so their training times are not directly comparable. The follow-up analysis E10 retrained both networks with fixed epochs and without early stopping (Section 6.5, Appendix F.2); its validation RMSEs are not comparable with these test results.

Mean RMSEs and the paired differences in A are reported in Section 7.1. In B the paired differences versus GlucoBench LR are −0.49 (95% CI −1.43 to 0.13) for the GRU and −0.51 (−1.42 to 0.13) for the TCN. Direct TCN-minus-GRU differences are 0.02 (−0.05 to 0.08) in A and −0.02 (−0.08 to 0.04) in B. The architectures give compatible exploratory results, with no supported winner between them.

A.3 Original-Paper PPGR Partial Reconstruction

The original paper reports a breakfast-level task that predicts the 2-hour area under the curve (AUC) and incremental area under the curve (iAUC) of the Libre response from anthropometric, biochemical and macronutrient predictors and the glucose value at the start of the meal. It uses an XGBoost model [4] (80 trees, maximum depth 1, learning rate 0.2, L1 regularization of 1, no L2 regularization) with leave-one-participant-out evaluation, and reports reference Pearson correlations of 0.89 (AUC) and 0.64 (iAUC). It does not fully specify the breakfast-inclusion rule, the treatment of missing or dictionary-flagged biomarker values, the HOMA-IR formula, the unit convention of the iAUC integral, or the random seed and software environment. The public author code supplies some of these choices (a HOMA-IR formula, integration

details, a meal-start baseline, train-only predictor scaling), but it also selects meal windows by row position without verifying timestamp continuity, aligns participant-level predictors by row position, and applies iAUC> 0 and first-breakfast exclusions that the paper does not state; these behaviors are evidence about that implementation, not specifications of the published experiment.

The reconstruction treats one eligible breakfast as one sample and predicts AUC and iAUC separately, using gap-free, timestamp-verified two-hour windows, trapezoidal integration in mg/dL·min, and iAUC as the positive area above the meal-start baseline. The 19 predictors are the energy from carbohydrate, protein, fat and fiber; meal-start glucose; age, gender and BMI; and the laboratory variables HbA1c, HOMA-IR, fasting insulin, fasting glucose and seven lipid measures. HOMA-IR is derived as fasting insulin times fasting glucose divided by 405, and HbA1c is used on the percent scale because the dictionary's mmol/mol label conflicts with the supplied values and with the original paper ($6.16 \pm 0.91\%$; p. 2, Table 2). Dictionary-designated error values are masked and complete cases are used; participant identity serves only to join participant-level data and to define the folds. Where the paper and the public code leave choices unresolved, these are stated reconstruction assumptions, so this is an evidence-bound partial reconstruction, not an exact reproduction.

On the complete-case primary cohort of 409 breakfasts from 43 participants, pooled Pearson correlations are 0.845 for AUC and 0.601 for iAUC (both $p < 0.001$), compared descriptively with the paper's 0.89 and 0.64; the difference is not attributed to a single implementation choice. Applying the public code's iAUC> 0 restriction leaves 403 meals and gives 0.867/0.602, and applying its additional nutrient scaling leaves the predictions unchanged. A positional join of static predictors would assign records to the wrong participant for 37 of the 43 participants in this release, so the reconstruction joins by participant ID; this identifies a risk in positional assignment and does not imply that the published analysis used an incorrect assignment.

B Models and Tuning

The factor-ablation pipeline fits one model for each registered configuration, cohort, evaluation protocol, and model family. Hyperparameters are selected by the lowest validation subject-mean RMSE; test data do not enter selection. The configuration identifiers below match the project's configuration registry (C01–C08 correspond to C1–C8 in the report).

B.1 Estimators, preprocessing, and candidate grids

Ridge uses `sklearn.linear_model.Ridge`, searched over $\alpha \in \{0.1, 1.0, 10.0\}$. The project pipeline first applies `SimpleImputer(strategy="median", keep_empty_features=True)` and then `StandardScaler`; both are fitted on training rows only. The core static inputs use their registered representations: raw numeric age and BMI and a male indicator with female as the reference level; no additional categorical encoding step is applied in the Ridge pipeline.

The GBT uses `sklearn.ensemble.HistGradientBoostingRegressor`, searched over the two combinations $(\texttt{learning_rate}, \texttt{max_depth}, \texttt{max_iter}) = (0.1, 3, 100)$ and $(0.1, 3, 300)$. The implementation sets `random_state=0` and `early_stopping=False`; it passes engineered inputs directly to the estimator without a project-level imputation or scaling pipeline.

B.2 Validation-selected values

Table 13 reports the selected hyperparameters recorded for each protocol and configuration. Configurations grouped in a row share the stated value; the GBT C08 Protocol B choice differs by cohort.
For every GBT row, the selected values of `learning_rate` and `max_depth` are 0.1 and 3, respectively. The values shown are those in the saved validation fits; no additional tuning was performed. Table 13 records all configurations fitted during model-family screening, whereas the test comparisons retain only the GBT configurations C01 and C04, as specified in Section 5.1.

B.3 Published-baseline implementation and adaptation

GlucoBench linear regression. A global OLS model with an intercept directly predicts the target from glucose lags at five-minute spacing [18]. Within the fixed 120-minute task history, candidate input lengths of

Table 13: Validation-selected factor-ablation hyperparameters.

Model	Protocol	Configuration / cohort	Selected value
Ridge	A	C01–C08, SC1, SC2, SC3 categorical and ordinal	$\alpha = 0.1$
Ridge	B	C01–C08, SC1, SC2, SC3 categorical	$\alpha = 0.1$
Ridge	B	SC3 ordinal (Gut cohort)	$\alpha = 10.0$
GBT	A	C01–C08, SC1, SC2, SC3 categorical and ordinal	`max_iter` = 300
GBT	B	C01–C03, C05	`max_iter` = 300
GBT	B	C04, C06–C07, SC1–SC2, SC3 categorical and ordinal	`max_iter` = 100
GBT	B	C08, core cohort	`max_iter` = 100
GBT	B	C08, full cohort	`max_iter` = 300

12 and 24 lags were compared using validation subject-equal RMSE; both protocols selected 24 lags for the test evaluation. No participant identifier or static covariate enters the model. This adaptation was checked against the public implementation, and the project's reimplementation, not the authors' reported numbers, is what is evaluated here.

Sparacino AR(1)-WLS. A local AR(1) coefficient is estimated at each origin by exponentially weighted least squares, with past pairs weighted by powers of a forgetting factor μ [21]. The 120-minute closed history window supplies 41 samples at three-minute spacing (40 consecutive regression pairs). With the fitted coefficient held fixed, future innovations are set to zero and the forecast is iterated for ten three-minute steps to produce the 30-minute-ahead point prediction. The project samples the released Libre grid and truncates fitting history to the task window. Unlike the original study, which applied a first-order Butterworth low-pass filter to attenuate artifacts in Glucoday measurements, this implementation uses the released, linearly interpolated Libre values without additional low-pass filtering. The candidates are $\mu \in \{0.2, 0.5, 0.8\}$, with validation subject-equal RMSE selecting $\mu = 0.2$ in both protocols. Closed-form WLS implements the weighted least-squares objective described in the original paper; no reference implementation was available to establish numerical equivalence with the original RLS routine.

B.4 Out-of-fold residuals and factor-block penalties

The training residuals used for screening, for OpenFE and for fitting the factor block are *out-of-fold*: in Protocol A each participant's training period is cut into five equal-duration blocks and the base predicting a block is fitted on the other blocks (windows whose 150-minute support crosses a block edge are dropped: 5,117 of 79,300 training windows, 6.45%, from the factor-block fit only; evaluation windows are unchanged); in Protocol B the base is fitted by five-fold grouped cross-validation over the 27 training participants. This is stricter than using in-sample fitted values (the baseline has only 25 parameters, so in-sample residuals were not expected to be biased), and was decided before the factor experiments. Validation and test windows use the base fitted on the full training split. The sequence ensembles of E10 use the same fold structure for their own out-of-fold predictions (Appendix F.2).

The ridge penalty α is chosen on validation data on a logarithmic grid from 10^{-3} to 10^{3} because the earlier ridge ablations selected the lower grid edge ($\alpha = 0.1$) almost everywhere and correlated candidates need regularization. The selected values for the primary configurations are at the grid edges ($\alpha = 1000$ for M in both protocols, $\alpha = 0.001$ for T and O); this is a limitation of the fixed grid and is discussed in Section 9.

C Supplementary Ridge Ablation and Error Analysis

This appendix reports a joint Ridge ablation of the manual feature groups and supplementary analyses of prediction errors. Unlike the main factor-discovery experiments, which evaluate additional factors against a fixed GlucoBench baseline through residual augmentation, the Ridge models jointly fit glucose history and the manual feature groups (activity A, diet D and core participant B added to the glucose-history features G; 42 paired contrasts). The analyses therefore address related but distinct questions, and their estimated feature increments are not directly comparable with those of Section 7. The contrasts are classified with the practical-effect rule of Section 6.3, and the test split had already been used in earlier project stages (Section 6.4). The

error strata are post hoc descriptions of the test predictions of these models; the concentration of the diet effect shortly after logged meals motivated the post-meal strata of Section 7.4, but this is not an independent confirmation.

C.1 Manual Feature-Group Ablation (Ridge on G)

Table 14 reports test performance for all eight core configurations; Figure 2 provides the paired-contrast overview across core and sensitivity analyses. Table 15 reports the core paired contrasts. The G-only reference has subject-equal RMSE of 11.30 mg/dL in A and 11.48 mg/dL in B.

Table 14: Test-set performance (mg/dL) for the eight core ablation configurations on the activity-availability-restricted common cohort.

		Subject-equal		Pooled-window	
Config	Feature groups	MAE	RMSE	MAE	RMSE
Protocol A: 44 participants, 21,083 test windows					
C1	G	7.82	11.30	7.74	11.44
C2	G+A	7.77	11.32	7.74	11.44
C3	G+A+D	7.72	11.29	7.68	11.40
C4	G+A+D+B	7.75	11.29	7.70	11.41
C5	G+D	7.74	11.26	7.68	11.40
C6	G+B	7.85	11.29	7.74	11.43
C7	G+D+B	7.78	11.25	7.68	11.40
C8	G+A+B	7.79	11.32	7.75	11.45
Protocol B: 9 participants, 26,861 test windows					
C1	G	7.70	11.48	7.71	11.52
C2	G+A	7.68	11.44	7.69	11.48
C3	G+A+D	7.49	11.17	7.51	11.22
C4	G+A+D+B	7.43	11.12	7.44	11.17
C5	G+D	7.49	11.17	7.50	11.22
C6	G+B	7.69	11.45	7.69	11.48
C7	G+D+B	7.48	11.14	7.49	11.19
C8	G+A+B	7.61	11.40	7.62	11.44

Diet is the only core group classified as improved under the pre-specified practical-effect rule with an interval excluding zero, and only in Protocol B. Its three contrasts reduce RMSE by approximately 2.4–2.6% relative to each contrast's own reference configuration. In Protocol A, all core intervals include zero. Activity intervals include zero in both protocols. Bio's two Protocol B intervals exclude zero, but their reductions are below the practical-effect magnitude threshold and remain classified as negligible; this is a small nonzero effect, not evidence of no association.

The validation pattern does not carry uniformly to test. All three Diet contrasts were classified as improved under the practical-effect rule in Protocol A validation (approximately -0.12 to -0.13 mg/dL), but none retained that classification on test. In Protocol B, only $G \rightarrow G$+D was classified as improved in validation; all three Diet contrasts were classified as improved on test. Across the 42 paired contrasts, validation and test practical-effect classifications differ for 11 of 21 contrasts in A and 5 of 21 in B.

Given this pattern, the Diet result is reported narrowly: adding dietary information produced a small, confidence-interval-supported reduction in test RMSE for the unseen-subject generalization protocol, and did not produce a confirmed reduction for the within-subject future-prediction protocol, despite an unconfirmed improvement at the validation stage in the latter case.

Table 15: Core-cohort subject-mean paired ΔRMSE [95% CI], mg/dL. Superscripts give the practical-effect class (N = negligible, I = improved; Section 6.3), not significance decisions.

Contrast	Protocol A	Protocol B
Activity		
$G \rightarrow G{+}A$	$0.019\,[-0.056, 0.107]^{N}$	$-0.035\,[-0.070, 0.003]^{N}$
$G{+}D{+}B \rightarrow G{+}A{+}D{+}B$	$0.046\,[-0.028, 0.131]^{N}$	$-0.018\,[-0.063, 0.030]^{N}$
Diet		
$G{+}A \rightarrow G{+}A{+}D$	$-0.032\,[-0.142, 0.070]^{N}$	$-0.271\,[-0.363, -0.174]^{I}$
$G \rightarrow G{+}D$	$-0.041\,[-0.148, 0.062]^{N}$	$-0.301\,[-0.394, -0.206]^{I}$
$G{+}A{+}B \rightarrow G{+}A{+}D{+}B$	$-0.028\,[-0.140, 0.077]^{N}$	$-0.274\,[-0.369, -0.177]^{I}$
Bio		
$G{+}A{+}D \rightarrow G{+}A{+}D{+}B$	$0.005\,[-0.050, 0.068]^{N}$	$-0.045\,[-0.075, -0.022]^{N}$
$G \rightarrow G{+}B$	$-0.012\,[-0.056, 0.041]^{N}$	$-0.029\,[-0.055, -0.008]^{N}$

C.2 Robustness and Sensitivity Analyses

Table 16 reports sensitivity configurations that qualify the core findings: SC1 delays meal availability by 30 minutes, SC2 adds offline laboratory values, and SC3 (Gut-health scores, Appendix A.1) is evaluated on its own reduced cohort. Delaying meal availability by 30 minutes worsens Protocol B performance by roughly the size of its immediate-Diet gain; the change in A is uncertain. Offline laboratory Bio has opposite point-estimate directions across protocols, but both intervals include zero, despite their practical-effect classifications as improved/worse.

Table 16: Sensitivity and model-structure subject-mean paired ΔRMSE [95% CI], mg/dL. Superscripts give the practical-effect class (N = negligible, I = improved, W = worse; Section 6.3), not significance decisions. For SC2 both intervals include zero, so the I/W classes are not interval-supported evidence of an effect in either direction.

Contrast	Protocol A	Protocol B
SC1/SC2: full feature set, core cohort		
Immediate → delayed Diet	$0.041\,[-0.063, 0.150]^{N}$	$0.284\,[0.186, 0.373]^{W}$
Full → full + labs	$-0.111\,[-0.234, 0.014]^{I}$	$0.186\,[-0.085, 0.492]^{W}$
Activity: all admitted windows		
$G \rightarrow G{+}A$	$0.243\,[0.102, 0.403]^{W}$	$-0.032\,[-0.064, 0.004]^{N}$
$G{+}D{+}B \rightarrow G{+}A{+}D{+}B$	$0.351\,[0.181, 0.539]^{W}$	$-0.014\,[-0.057, 0.032]^{N}$
Model structure: Ridge → shallow GBT, core cohort		
G only	$1.910\,[1.490, 2.319]^{W}$	$1.908\,[0.631, 2.959]^{W}$
$G{+}A{+}D{+}B$	$1.514\,[1.064, 1.960]^{W}$	$2.151\,[1.032, 3.195]^{W}$

On all admitted windows, both Activity contrasts worsen Protocol A performance with intervals excluding zero; Protocol B remains negligible. This differs from the core-cohort Activity result and cannot be generalized across cohorts. Shallow GBT underperforms Ridge in all four matched configuration/protocol comparisons, with every interval excluding zero. Figure 2 shows all 42 contrasts.

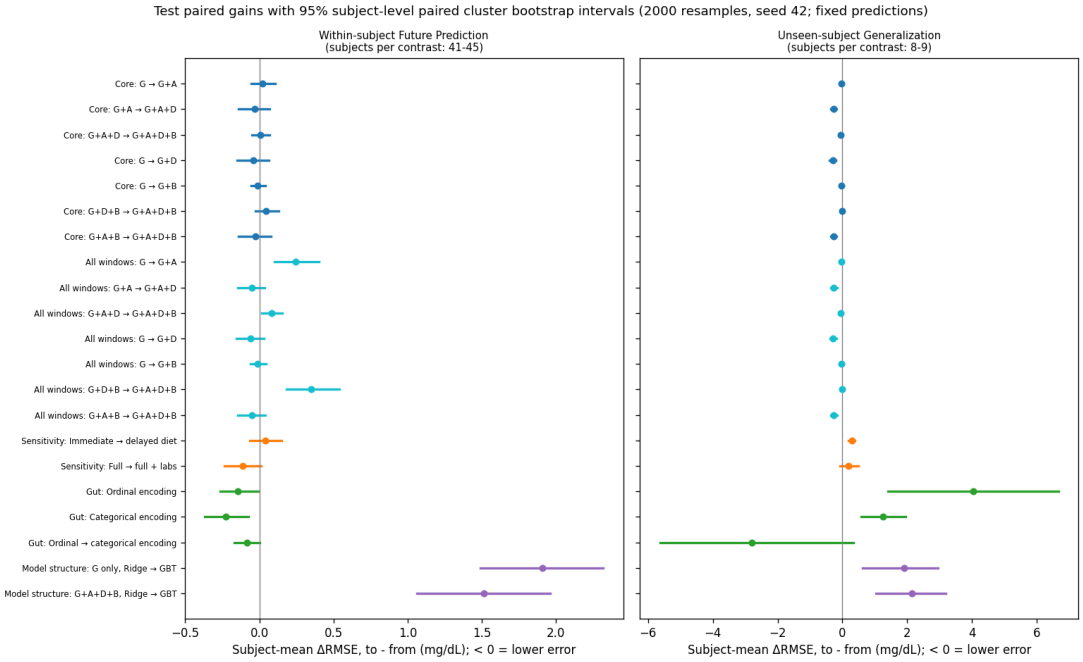


Figure 2: All 42 paired ΔRMSE contrasts with 95% subject-cluster bootstrap intervals (negative values favor the destination configuration; scales differ between protocols). Core contrasts are in Table 15, sensitivity and model-structure contrasts in Table 16, and the Gut-score contrasts use their own reduced cohort (Appendix A.1).

C.3 Error Analysis of the Ridge Ablation

The following strata describe the test predictions of these models post hoc. Realized glucose change, meals logged after the origin, and Dexcom at the target time use post-origin information and are diagnostic strata, not model inputs or confirmatory tests.

C.3.1 Error by Glucose Level and Realized Change

Figure 3 shows increasing error at high target glucose and during rapid changes. Ridge configurations behave similarly and generally improve on persistence, although persistence is competitive in the lowest glucose band. Per-participant error also varies with glucose variability, without establishing a feature-specific relationship.

C.3.2 Diet-Associated Error Stratification

Diet-associated reductions are concentrated in the first 30 minutes after a logged meal: for $G{+}D$ versus G, pooled ΔRMSE is -1.97 mg/dL in B and -0.72 mg/dL in A. Other meal-time strata show small or inconsistent differences. Meals logged only after the origin provide no gain; this localization is consistent with the delayed-diet sensitivity (SC1).

C.3.3 Floor Censoring and Extreme Errors

The Libre record has a floor of 40 mg/dL. Across the two protocol evaluations, 2,130 test targets lie at this floor, concentrated in a few participants; these are evaluation counts, not independent observations across protocols. On core-cohort floor windows in B, Ridge and GlucoBench LR have positive bias of approximately 3.7–4.6 mg/dL against the recorded target. The latent glucose value below a censored reading is unobserved, so error against that latent value cannot be recovered or its direction inferred from the aggregate recorded-target bias.

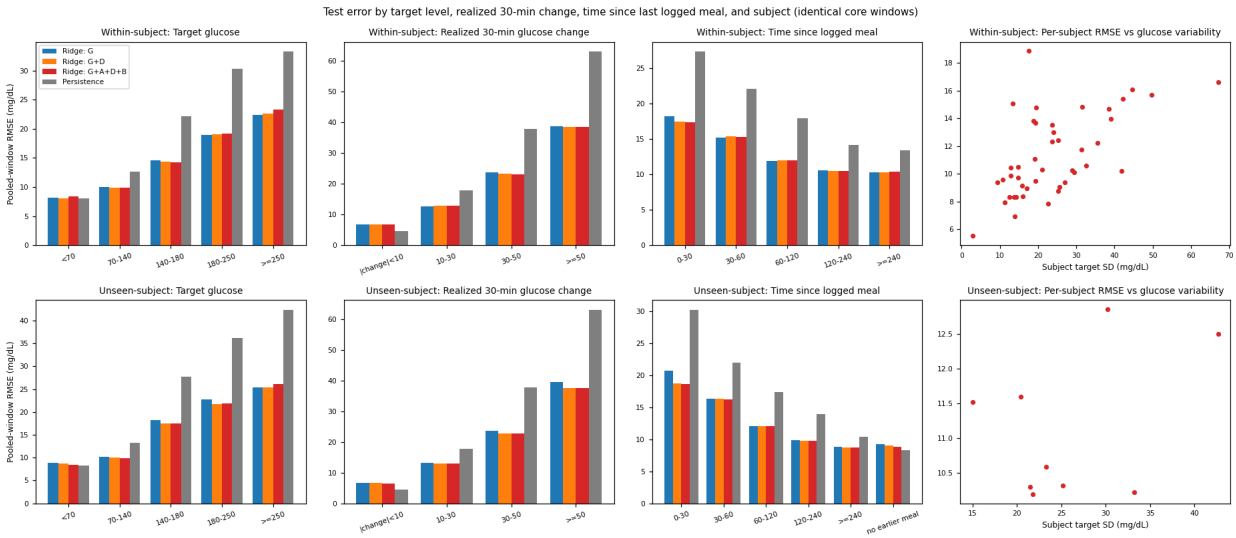


Figure 3: Pooled-window test RMSE stratified by target glucose level, realized 30-minute change, and time since the most recently logged meal, for persistence and three core-cohort Ridge configurations, on identical windows within each protocol. The rightmost panels show per-participant RMSE against that participant's own test-window glucose standard deviation.

Unconstrained linear predictions occasionally become negative in B (4 for each of the three core Ridge configurations shown, 11 for GlucoBench LR), whereas none do so in A; the models inspected in this diagnostic have no predictions above 400 mg/dL. Nine of the ten largest absolute-error cases in Protocol B occur in one participant and coincide with large Libre–Dexcom disagreements across several glucose-based models. The cause of these discrepancies remains uncertain. Across the broader set of windows, Libre–Dexcom disagreement does not show a consistent monotonic relationship with prediction error.

D Definitions, Factor Grammar and Selected-Factor Catalogue

D.1 Feature groups, factor roles and vocabulary

This subsection collects the definitions to which Sections 3 and 4 refer.

Manual feature groups.

G: **glucose history.** Ten features: glucose at lags 0, 30, 60, and 120 minutes, plus mean, population standard deviation, and OLS slope with intercept over each of the closed 30- and 120-minute trailing windows.

A: **activity.** Over both 30- and 120-minute windows, heart rate, activity calories, and MET contribute observed-value means, maxima, coverage fractions, and missingness indicators; activity calories also contribute observed sums. MET and the alternative Intensity schema are not interchangeable. Missing activity is not coded as rest or zero.

D: **diet.** Over both 30- and 120-minute windows, logged-meal counts, no-logged-meal flags, and sums of calories, carbohydrate, protein, fat, and fiber are computed. The most recent admissible meal within 120 minutes supplies elapsed time, calories, carbohydrate, and meal-type indicators. Nutrient totals are missing when no meal is logged or a contributing nutrient is missing, rather than assuming zero intake.

B: **core Bio.** Age, self-reported gender, and BMI, available for all 45 participants and measured at intake. Laboratory values were taken on the first day but their return time is not reported, so they enter only offline extensions. HbA1c is excluded from the forecasting configurations (Table 17); its separate PPGR use is documented in Appendix A.3.

Table 17: Factor roles and prediction-time availability of candidate data sources.

Candidate source	Training-scope / availability consideration	Final role
Recent glucose	Dense primary history signal; available at prediction time under the task definition.	Core *G*
Heart rate / activity calories / MET	Time-varying activity; incomplete coverage is represented explicitly rather than coded as rest.	Core *A*
Meal logs / macronutrients	Recent dietary information; logged timestamps are used under an explicit availability assumption.	Core *D*
Age / gender / BMI	Static participant information, available for all participants.	Core *B*
Laboratory variables	Baseline covariates measured at screening on day 1; the time at which results were returned is not reported.	Offline extension (SC2) of the Ridge ablation (Appendix C); the meal-by-laboratory-state interaction family was not generated for the formal comparisons and was evaluated afterwards on validation data only (E11, Appendix F.3)
Gut-health scores	Partial participant coverage.	Restricted-cohort sensitivity
Microbiome	Partial coverage and high-dimensional subject-level data.	Exploratory extension
Meal photographs	Logged timestamps are used as meal-event times under an availability assumption; photograph-derived features and post-meal consumed-proportion estimates are excluded.	Content excluded
HbA1c	Percent scale (original paper, p. 2, Table 2: $6.16 \pm 0.91\%$; data range 4.6–8.5); the mmol/mol label in the data dictionary is a dictionary error.	Not used in the Ridge ablation (decided before the unit was corrected and kept); used as a baseline covariate in the PPGR reconstruction

D.2 Factor Grammar Specification

Under the current operator whitelist, primitive inputs are restricted to values at or before the origin t. Earliest and latest source offsets are propagated recursively through nested candidate expressions; candidates are rejected when the latest offset is positive or the earliest offset extends beyond the allowed history window. The latest-offset check relies on the temporal semantics of the permitted primitives and is not an automatic provenance guarantee for arbitrary future operators. Candidate sizes were fixed before any search (206 in the main grid; the laboratory family of 12 more was not generated for T and is evaluated in E11, Appendix F.3). Table 19 lists the families, operators and grids.

The time-to-peak τ^* and shape k^* of each nutrient and meal type used by the interaction and ratio families are chosen on the training data as the pair with the largest absolute mean subject-level Spearman correlation with the out-of-fold residual (the signed mean and the same-sign share are recorded). Anchor rule: period 15 minutes, tolerance 10^{-6} for a kink, at least four supporting kinks, support over eight kinks, phase share at least 0.9, maximum staleness 60 minutes, at most 8 anchors (9 for the anchor-causal base of E8).

Table 18: Factor-discovery terminology: concepts, interpretation and limitations.

Concept	Wording	Can answer	Cannot answer
Manual features	domain-designed transformations of raw inputs	whether an expert-chosen representation is useful	unforeseen combinations
Automated construction	algorithmic generation of candidate transformations from operators or programs	whether a better representation exists in a search space	the scientific meaning of a factor
Feature selection	choosing a subset under a criterion	which candidates help under that criterion	causation; reliability when selection is unstable
Association (training)	subject-level rank correlation with the baseline residual	whether a dependence exists in training data	out-of-sample usefulness
Predictive utility	paired held-out error reduction over the fixed baseline	whether a block improves prediction given the baseline	mechanism
Stability	selection frequency under subject subsampling; consistency across groups	whether a selection is robust across participants	external populations; a proven error rate
Interpretability	explicit formula, sign and dose–response	whether a factor can be read	whether the reading is correct
Causal effect	interventional change	–	not addressed by this data and design

Table 19: Theory-constrained grammar T: families, operators and grids.

Family	Operator	Grid	Count
Meal absorption	`conv_gamma`: pulses of one nutrient of one meal type convolved with a gamma kernel of unit sum over ages 0–120 min; τ = time to peak	$\tau \in \{20, 30, 45, 60, 90\}$ min; $k \in \{2, 3\}$; nutrients carbohydrate, protein, fat, fiber; meal types breakfast, lunch, dinner, snack	160
Meal × baseline state	kernel-weighted carbohydrate of a meal type × state (first-order interaction); (τ, k) of the carbohydrate of that meal type	state ∈ {BMI, last native reading, native slope over 1 h}; four meal types	12
Nutrient ratios	carbohydrate/(carbohydrate+protein+fat); fiber/carbohydrate; protein×carbohydrate	last meal or kernel-weighted totals	6
Activity kernels	`ts_mean`, `ts_max` of heart rate and activity calories; exponential decay; activity × meal kernel	windows 15, 30, 60, 120 min; decay $\tau \in \{30, 60\}$ min	24
Native glucose state	last inferred native reading; difference of the last two; slope and curvature over the last hour	at most 8 anchors	4
Main grid			206
Laboratory × meal (not generated for T; see E11)	kernel carbohydrate × {fasting glucose, HbA1c (%), HOMA-IR}	four meal types	(12)

D.3 Selected Factor Catalogue

Table 20 gives the selected sets in the order of forward selection; Table 21 gives the training-scope screening statistics of the selected T_k members, that is, their association with the baseline residual on which candidates were ranked before validation selection (descriptive; association is not predictive utility). The half-sample frequency of the selected and most stable factors is in Table 34, and all candidates with their formulas, units, source times and generation status are in the project repository. Each block also carries a missing indicator for windows whose native-reading state is uncertain.

E OpenFE and LLM-Guided Search: Configuration

OpenFE. Official library 0.0.12, unmodified, in a separate environment (scikit-learn 1.5.2, LightGBM 4.7.0); settings are the library defaults except feature boosting; stage-one metric predictive, stage-two gain importance, eight data blocks, at least 2,000 candidate features, seed 42, windows down-sampled to a 15-minute step. Initial scores are the baseline's out-of-fold predictions; the internal validation is the last 25%

Table 20: Selected factor sets of the theory-constrained constructor (T_k) and of OpenFE (O_k); formulas of O use the project's primitive names.

Set	Protocol	k	Factors in the order of forward selection (validation-only selection)
T_k	A	6	actstat_calories_mean_15m; int_snack_carbs_x_native_slope_1h; int_lunch_carbs_x_bmi; actstat_hr_mean_120m; actstat_hr_max_15m; int_breakfast_carbs_x_native_slope_1h
T_k	B	7	actstat_hr_max_15m; abs_lunch_carbs_t20_k2; int_snack_carbs_x_native_slope_1h; ratio_last_meal_carb_share; ratio_kernel_protein_x_carbs; int_dinner_carbs_x_native_slope_1h; abs_dinner_fiber_t20_k3
O_k	A	7	(diet_no_logged_meal_30m+native_curv_1h); (diet_minutes_since_last_logged_meal/bio_age); (diet_last_meal_type_lunch-native_slope_1h); (diet_no_logged_meal_30m*act_calories_mean_15m); (diet_no_logged_meal_30m*act_met_mean_30m_lag10); (act_calories_mean_60m*meal_any_logged_minutes_since); (diet_logged_meal_count_30m+native_curv_1h)
O_k	B	4	(diet_minutes_since_last_logged_meal+native_diff); (act_calories_mean_15m/act_calories_mean_60m); (meal_breakfast_logged_minutes_since*native_last); (diet_no_logged_meal_30m/native_last)

Table 21: Training-scope screening statistics of the selected T_k members: mean subject-level Spearman information coefficient (IC) with the out-of-fold baseline residual, share of subjects with the same sign, number of subjects and rank within the family (training windows only; descriptive, association is not predictive utility).

Protocol	Factor	Mean IC	Same-sign share	Subjects	Rank in family
A	actstat_calories_mean_15m	+0.067	0.89	44	1
A	int_snack_carbs_x_native_slope_1h	-0.031	0.71	35	2
A	int_lunch_carbs_x_bmi	+0.031	0.68	44	3
A	actstat_hr_mean_120m	-0.046	0.82	45	5
A	actstat_hr_max_15m	+0.047	0.80	45	4
A	int_breakfast_carbs_x_native_slope_1h	-0.039	0.70	44	1
B	actstat_hr_max_15m	+0.061	0.93	27	3
B	abs_lunch_carbs_t20_k2	+0.025	0.59	27	1
B	int_snack_carbs_x_native_slope_1h	-0.041	0.80	20	1
B	ratio_last_meal_carb_share	+0.039	0.70	27	2
B	ratio_kernel_protein_x_carbs	+0.044	0.59	27	1
B	int_dinner_carbs_x_native_slope_1h	-0.037	0.69	26	2
B	abs_dinner_fiber_t20_k3	+0.025	0.62	26	2

of each training period (A) or an 18/9 grouped split of the 27 training participants (B). Inputs: the manual features excluding the eight unlagged MET features, non-glucose primitives (including MET inputs lagged by at least ten minutes), baseline covariates admitted by the decision in the frozen protocol, and the native-reading state; no interpolated lags and no subject column. The first 25 ranked formulas of each protocol enter the same redundancy filter and forward selection as T (pools of 23 in A and 21 in B after the filter; $k = 7$ and 4). The formulas are applied row by row by project code and require no fitted cross-row statistics; OpenFE's own transform is not called. Because the OpenFE inputs include manual features, O may re-use them.

LLM-guided search (L) and Random-DSL (R). Candidates were proposed by Claude Sonnet 5 (`claude-sonnet-5`) through the Claude Code command-line client in non-interactive mode, with tools disabled and no fallback model. Sampling parameters were not set, so repeated calls are not guaranteed to return identical proposals; the seed 42 of the configuration controls only the memory sampling and Random-DSL, not the model. The search design, budget and evaluation split are described in Section 4.4. Each branch (called an island in the implementation and saved logs) keeps its own memory of its five best expressions; each prompt includes up to two of them (none in the first round), sampled with a selection temperature of 0.01, together with their rounded training-fold gains and recent rejection reasons. The prompt states the task and the operator list, requires one hypothesis sentence per proposal, and contains no data rows: a guard rejects timestamps, subject identifiers and numbers other than rounded gains. Unparsable replies count as calls without proposals, and an expression already evaluated in any branch counts as a proposal but

is not evaluated again. Neither search was evaluated on test data. The stored requests and replies allow the recorded search to be replayed and audited; they do not guarantee that re-querying the model reproduces the same replies.

Selected L and R factors. Table 22 lists the 11 selected L factors with the hypotheses exactly as written by the model (a motivation, not a finding) and Table 23 the 12 selected R factors; the order is the selection order on validation data, the rank is the position of the expression by training fitness among the unique evaluations of its search, and the nearest factor of T or O is the one with the highest absolute Spearman correlation with it (only T and O were searched; there is no such analysis for M).

Table 22: Selected factors of the LLM-guided search (L): expression, category, model-stated hypothesis (verbatim), training fitness (rank in the search) and the nearest factor of T or O with the signed Spearman correlation. IDs L_A_ and L_B_: the first six rows are Protocol A, the last five Protocol B.

ID	Expression	Category	Model-stated hypothesis (verbatim)	Train. fitness (rank)	Nearest T/O (ρ)
L_A_02	`ratio(ts_mean("hr", 30), ts_mean("hr", 120))`	activity	A recent heart-rate elevation relative to the longer-term baseline may indicate acute physical activity that transiently suppresses glucose rise.	0.0390 (2)	T +0.39
L_A_19	`interact(conv_gamma("any", "carbs", 30, 2), native("slope_1h"))`	meal × native glucose state	The combination of current glucose momentum and recent carbohydrate load should capture nonlinear acceleration effects not represented in a fixed linear baseline.	0.0101 (19)	O −0.70
L_A_01	`ratio(conv_gamma("any", "carbs", 20, 2), ts_mean("hr", 30))`	activity × meal	Fast-absorbing carbohydrate exposure may translate into a smaller glucose rise when recent heart rate (a proxy for ongoing physical activity) is elevated, since exercise increases glucose uptake.	0.0425 (1)	O −0.55
L_A_06	`interact(last_meal("carbs"), minutes_since_meal("any"))`	meal	The glycemic impact of the most recent meal's carbohydrate load likely depends on how long ago it was eaten, since absorption dynamics evolve over time.	0.0190 (6)	O +0.60
L_A_09	`ratio(conv_gamma("any", "carbs", 30, 2), conv_gamma("any", "fat", 60, 2))`	meal	The relative weight of fast-absorbing carbohydrates versus slower-absorbing fat intake should indicate how quickly glucose is likely to rise, improving residual prediction.	0.0173 (9)	O −0.51
L_A_21	`ratio(conv_gamma("lunch", "carbs", 45, 2), native("last"))`	meal × native glucose state	The expected glycemic rise from a lunch carbohydrate load may be proportionally smaller when the current glucose level is already high, reflecting saturation of absorption effects.	0.0082 (21)	T +1.00 [near-duplicate]
L_B_01	`ratio(last_meal("carbs"), minutes_since_meal("any"))`	meal	The carbohydrate content of the most recent meal should have a diminishing residual effect as more minutes elapse since it was logged, so normalizing by elapsed time may isolate its decaying influence	0.0474 (1)	T +0.75
L_B_02	`interact(minutes_since_meal("any"), ts_mean("hr", 30))`	activity × meal	Elevated heart rate following a meal reflects a postprandial thermic/metabolic response that, combined with elapsed time since eating, may explain additional variance in glucose trajectory.	0.0294 (2)	O +0.91
L_B_03	`ts_max("hr", 30)`	activity	A recent peak in heart rate may indicate acute physical exertion that transiently alters glucose utilization and thus the forecast residual.	−0.0007 (3)	T +0.94
L_B_09	`interact(conv_gamma("any", "carbs", 30, 2), native("slope_1h"))`	meal × native glucose state	Fast-absorbing recent carbohydrate load should amplify or dampen an existing glucose trend, so their product may explain residual curvature in the forecast.	−0.0087 (9)	T +0.51
L_B_08	`ratio(last_meal("carbs"), bmi())`	BMI (subject-level) × meal	The glycemic impact of a given carbohydrate load may scale inversely with body size, so normalizing the most recent meal's carbs by BMI could better predict the residual.	−0.0082 (8)	T +0.40

Table 23: Selected factors of the random control (R): expression, training fitness (rank in the search) and nearest factor of T or O with the signed Spearman correlation; the first four rows are Protocol A, the last eight Protocol B.

ID	Expression	Category	Train. fitness (rank)	Nearest T/O (ρ)
R_A_02	`ratio(exp_decay("hr", 60), exp_decay("hr", 30))`	activity	0.0577 (2)	T −0.41
R_A_01	`minutes_since_meal("breakfast")`	meal	0.0959 (1)	O +0.92
R_A_09	`interact(native("slope_1h"), nutrient_ratio("protein_x_carbs"))`	meal × native glucose state	0.0059 (9)	O −0.63
R_A_03	`ratio(last_meal("carbs"), native("last"))`	meal × native glucose state	0.0240 (3)	T +0.27
R_B_02	`minutes_since_meal("breakfast")`	meal	0.0306 (2)	O +0.89
R_B_04	`ratio(last_meal("carbs"), native("last"))`	meal × native glucose state	0.0093 (4)	T +0.42
R_B_03	`interact(ts_max("calories", 15), ts_max("hr", 60))`	activity	0.0113 (3)	T +0.86
R_B_24	`ratio(exp_decay("calories", 30), ts_mean("hr", 15))`	activity	−0.0095 (24)	T +0.54
R_B_15	`interact(native("slope_1h"), nutrient_ratio("protein_x_carbs"))`	meal × native glucose state	−0.0082 (15)	T +0.38
R_B_11	`minutes_since_meal("dinner")`	meal	−0.0064 (11)	O +0.91
R_B_09	`interact(nutrient_ratio("protein_x_carbs"), ts_max("calories", 15))`	activity × meal	−0.0054 (9)	T +0.47
R_B_13	`interact(nutrient_ratio("carb_share"), ts_max("hr", 15))`	activity × meal	−0.0071 (13)	T +0.91

F Validation-Only and Exploratory Analyses

The analyses in this appendix use training or validation data only and give optimistic estimates; they are not test results. Appendix F.2 and Appendix F.3 give the details and complete results of the follow-up analyses E10 and E11 of Section 6.5. The one exception is the descriptive test comparison by HbA1c group cited below, which is taken from Appendix G (Table 36).

F.1 Original Validation-Only Analyses

Metabolic groups as environments. Within the training participants of each protocol, the three HbA1c groups were treated as environments, and a factor was called consistent if the sign of its mean information coefficient and of its group-wise ridge coefficient was the same in all groups (no factor was reselected): 4 of the 6 members of T_k in A and 2 of 7 in B. This is an analogy to stabilized regression [17], not an implementation of it. The test results by HbA1c group (Table 36 in Appendix G; three participants per group in B, 14–16 in A) show no stable gradient across groups.

LLM-guided search versus random search (E3): further validation contrasts. The validation contrasts of L and R against the baseline and against each other, and the search statistics, are in Table 8 (Section 7.6). Table 24 gives the remaining contrasts of the same experiment, including both sets added on top of M, as stored in the validation file; they are validation estimates and optimistic.

Table 24: E3, further validation contrasts: ΔRMSE [95% CI], mg/dL (negative: the second configuration is better). Protocol A uses the 44 participants with at least 20 validation windows.

Contrast	Protocol A ($n = 44$)	Protocol B ($n = 9$)
+T → +L	−0.010 [−0.062, +0.046]	−0.074 [−0.150, +0.003]
+T → +R	−0.044 [−0.109, +0.021]	−0.086 [−0.173, −0.005]
+O → +R	+0.039 [−0.022, +0.101]	+0.056 [−0.000, +0.104]
+M → +M+L	−0.110 [−0.167, −0.062]	−0.008 [−0.066, +0.049]
+M → +M+R	−0.057 [−0.085, −0.029]	−0.039 [−0.065, −0.008]
+M+R → +M+L	−0.053 [−0.115, −0.001]	+0.031 [−0.032, +0.097]

Selection stability (E4). Fifty complementary pairs of half-samples of the training participants (22/23 of 45 in A, 13/14 of 27 in B), the same screening and an L1 path on the residual as selector (the first penalty

with at least ten non-zero coefficients; the largest coefficient defines the choice). Stable factors at thresholds 0.5/0.6/0.7: A 8/7/4, B 7/7/5. Within-pair Jaccard of the selected sets: 0.29 (A), 0.30 (B); leave-one-subject-out Jaccard: 0.75 (A); Jaccard between the full-sample selections of A and B: 0.54. Frequencies are descriptive and not error rates.

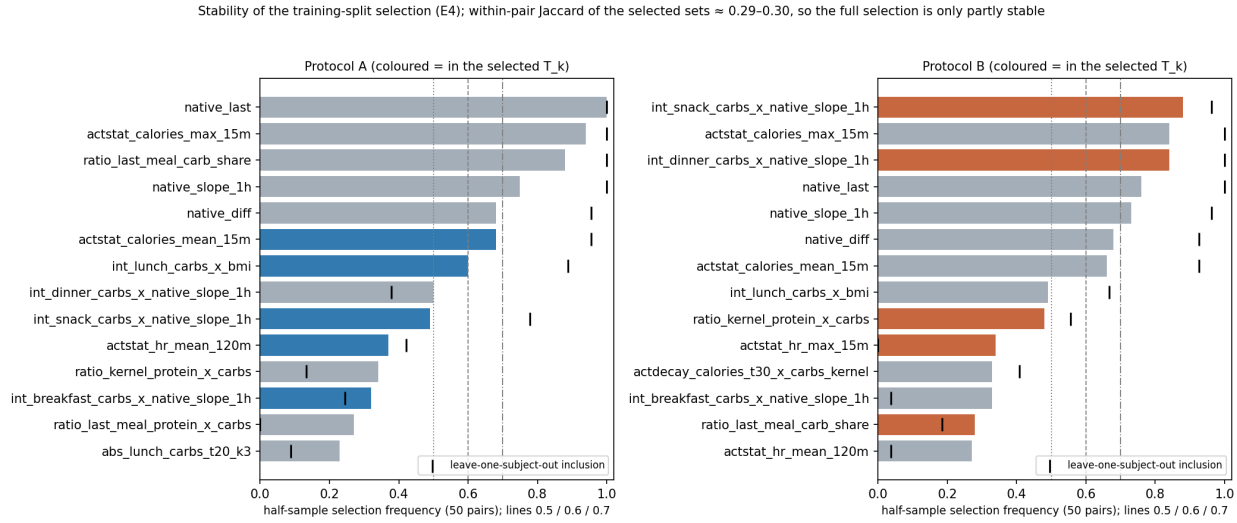


Figure 4: Half-sample selection frequency of the most frequently selected candidates (E4; 50 complementary pairs of training participants) and leave-one-subject-out inclusion, by protocol. Colored bars are members of T_k; vertical lines mark the thresholds 0.5, 0.6 and 0.7. Frequencies are descriptive, not error rates.

Target-form and single-feature checks. For a gradient-boosted model, predicting the change $y - g(t)$ rather than the level changes validation RMSE by -0.083 (A) and -0.100 (B), negligible, and the boosted model stays worse than the linear base. Of the 50 manual features, the largest single-feature additions to the base are diet_minutes_since_last_logged_meal (-0.090) in A and diet_no_logged_meal_30m (-0.124) in B; two additions reach 0.1 mg/dL in B and none in A, and no single removal from M reaches 0.1.

Individual breakfast response. Interacting an individual carbohydrate peak per gram from repeated breakfasts with the absorption kernel changed validation RMSE by less than 0.01 mg/dL in A.

Subject-identity proxy. The subject-identity proxy (E6) is reported in Section 7.7.

F.2 Frozen-Factor Transfer to Sequence Baselines (E10)

The network architectures, the optimizer (Adam, no weight decay), the loss (mean squared error of the standardized target-minus-latest-glucose residual), the seeds 0–4 and the training statistics follow Appendix A.2.2; the epochs are fixed as described in Section 6.5. For each network and protocol, five OOF folds and one full-training fit were run for each of the five seeds (30 fits; 120 in all); the GRU was trained on a GPU and the TCN on a CPU. In A, the five OOF blocks are the original non-forward temporal blocks within each participant, so the windows whose support crosses a block edge (Table 25) have no OOF prediction; in B the folds are the original five participant-grouped folds, with held-out participants excluded from the corresponding fit. The Ridge label is the OOF residual of the ensemble, and imputation, standardization and coefficients use training rows only. The penalty α is chosen on the grid $10^{-3}, \ldots, 10^{3}$ by the lowest subject-equal RMSE on the core validation windows (all nine participants of B meet the 20-window rule), and LR, ensemble and the three block models are compared on exactly the same windows (Table 27 lists their RMSEs and the selected penalties). Validation predictions come from models fitted on the full training

split only, without a training-plus-validation refit. The blocks are the frozen sets M, T_k and O_k with their missing-state indicator columns.

Table 25: E10: training windows, legal out-of-fold (OOF) windows and validation windows by protocol. Windows whose 150-minute support crosses an OOF block edge have no OOF prediction and are excluded from the OOF models and the Ridge fit in A; the validation windows are not affected by this exclusion.

Prot.	Training	Legal OOF	Excluded	OOF coverage	Validation (admitted)	Reported (n / windows)
A	79,300	74,183	5,117	93.55%	26,330	44 / 26,300
B	81,958	81,958	0	100.00%	24,259	9 / 23,139

Table 26: E10: GlucoBench LR and the fixed sequence ensembles on the shared validation windows (subject-equal RMSE in mg/dL); ΔRMSE = sequence − LR [95% CI], Impr. = share of participants with lower error. The comparison mixes architecture, input representation and training procedure.

Network	Prot.	n	Windows	LR	Sequence	ΔRMSE [95% CI]	Impr.
GRU	A	44	26,300	11.090	10.341	−0.748 [−1.325, −0.393]	89%
GRU	B	9	23,139	12.468	11.844	−0.624 [−1.071, −0.283]	100%
TCN	A	44	26,300	11.090	10.355	−0.735 [−1.303, −0.379]	84%
TCN	B	9	23,139	12.468	11.781	−0.688 [−1.149, −0.374]	100%

Table 27: E10: subject-equal RMSE (mg/dL) of the sequence ensemble and of the ensemble with each frozen block, on the shared validation windows, and the Ridge penalties α chosen on validation data (M/T_k/O_k).

Network	Prot.	Sequence	+M	+T_k	+O_k	α
GRU	A	10.341	10.227	10.240	10.179	1000/1000/0.001
GRU	B	11.844	11.719	11.743	11.688	1000/1000/1000
TCN	A	10.355	10.247	10.269	10.225	1000/1000/0.001
TCN	B	11.781	11.657	11.697	11.646	1000/1000/1000

Table 28: E10: subject-equal validation RMSE (mg/dL) of the single seeds, their mean and the ensemble (per-window mean of the five predictions), shared validation windows. No seed was selected.

Network	Prot.	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Mean	Ensemble
GRU	A	10.472	10.427	10.448	10.432	10.295	10.415	10.341
GRU	B	11.933	11.877	11.922	12.005	11.880	11.923	11.844
TCN	A	10.545	10.378	10.471	10.405	10.450	10.450	10.355
TCN	B	11.949	11.889	11.978	11.872	11.819	11.901	11.781

In Table 28, the ensemble RMSE is lower than the seed mean in all four settings and lower than every single seed except seed 4 of the GRU in A, which is a property of averaging predictions. No seed was selected on the basis of validation performance, and the fixed epoch counts were inherited from historical validation-based sequence checks rather than retuned in E10. In A the GRU ensemble is slightly better than the TCN ensemble and in B the order is reversed; the two networks were not compared with a paired test. The caveats of Sections 6.5 and 9 apply. The analysis adds a factor block to a fixed network by residual augmentation; it does not train the factors jointly with the network.

F.3 Offline Laboratory-State Interactions (E11)

Design and samples. The notation of Section 6.5 applies. The kernel parameters (τ, k) are those of the original training-scope records and were not reselected: breakfast $(90, 3)$, lunch $(20, 2)$, dinner $(20, 3)$ and snack $(20, 3)$ in A, and $(20, 2)$ for all four in B. Each construction is followed by the usual block standardization on the training windows; C-S1 uses equal weights per training participant and the population standard deviation. The identity proxy of a training participant is the mean of their training OOF residuals and is not estimated from validation data. The three laboratory variables are complete for all participants.

The training samples are 45 participants and 74,183 core OOF windows in A and 27 and 79,294 in B; the validation samples are 44 participants and 26,300 windows in A (one participant with a single window falls under the 20-window rule) and nine and 23,139 in B. The penalty of each of the seven nested or non-nested configurations is chosen on the same validation windows by the lowest subject-equal RMSE, and the intervals use the same bootstrap as the formal comparisons. Two checks preceded the analysis: the original E6 result was reproduced exactly (A, at least 20 validation windows: -0.0785 [-0.152, -0.027] without and -0.0002 [-0.0014, 0.0009] with the proxy, within 10^{-8} mg/dL), and the BMI interactions of the formal comparisons were reproduced exactly (maximum difference zero). The three laboratory columns of E11 are not those of E6, so E11 values need not match E6.

Table 29: E11: main comparison $R_{\text{ref}}+K_{\text{meal}}+L_{\text{lab}} \rightarrow +X_{\text{int}}$ on validation data (core cohort, at least 20 windows per participant; subject-equal RMSE in mg/dL). C-S1: laboratory states standardized on the training participants; C-S2: unstandardized states; proxy: training identity proxy added in A. ΔRMSE [95% CI] = with X_{int} − without; Impr. = share of participants with lower error. Exploratory, optimistic validation estimates.

Setting	n	Windows	Without X	With X	ΔRMSE [95% CI]	Impr.
A, C-S1	44	26,300	10.796	10.777	−0.019 [−0.052, +0.013]	59%
A, C-S2	44	26,300	10.796	10.779	−0.017 [−0.051, +0.013]	64%
A, C-S1, proxy	44	26,300	10.772	10.752	−0.019 [−0.053, +0.012]	64%
A, C-S2, proxy	44	26,300	10.772	10.755	−0.017 [−0.051, +0.015]	59%
B, C-S1	9	23,139	12.188	12.166	−0.023 [−0.076, +0.030]	67%
B, C-S2	9	23,139	12.188	12.162	−0.026 [−0.065, +0.009]	67%

Table 30: E11: further validation comparisons in the primary construction (C-S1): ΔRMSE [95% CI], mg/dL. Ref = fixed GlucoBench prediction plus the residual block of M; K = four meal kernels; Lab = three laboratory states; X = twelve interactions. The last two rows are not hierarchical or not nested (the first adds interactions without the additive laboratory states). Same participants and windows as Table 29.

Comparison	A	A, proxy	B
Ref → Ref+Lab	−0.080 [−0.157, −0.027]	+0.0003 [−0.0008, +0.0014]	−0.087 [−0.241, +0.059]
Ref → Ref+K	−0.063 [−0.116, −0.023]	−0.060 [−0.109, −0.021]	−0.030 [−0.052, −0.007]
Ref+K → Ref+K+Lab	−0.077 [−0.153, −0.024]	+0.0003 [−0.0007, +0.0013]	−0.087 [−0.240, +0.059]
Ref+K → Ref+K+X	−0.039 [−0.088, +0.005]	−0.019 [−0.052, +0.012]	−0.072 [−0.162, +0.003]
Ref+Lab → Ref+X	+0.008 [−0.063, +0.079]	−0.045 [−0.092, −0.006]	−0.004 [−0.115, +0.115]
Ref+K+Lab → Ref+K+X	+0.038 [−0.019, +0.103]	−0.019 [−0.053, +0.011]	+0.015 [−0.099, +0.134]

Hierarchy of the comparisons. Table 30 separates the contribution of the additive terms from that of the interactions. In A without the proxy, adding L_{lab} to the reference or to the kernels lowers the error by about 0.08 mg/dL; with the proxy the same change is practically zero, so the additive effect is absorbed by the proxy, as in E6. The interactions added to the kernels alone give a larger point estimate than when the additive states are present (A without the proxy, and B), so part of that gain is not an interaction effect. The non-nested replacement of the additive states by the interactions ($R_{\text{ref}}+K_{\text{meal}}+L_{\text{lab}} \rightarrow R_{\text{ref}}+K_{\text{meal}}+X_{\text{int}}$) shows no advantage of interactions over additive terms in either protocol: all its intervals include zero. A further comparison does not isolate the interactions: replacing L_{lab} by X_{int} in the reference alone ($R_{\text{ref}}+L_{\text{lab}} \rightarrow R_{\text{ref}}+X_{\text{int}}$) gives -0.045 [-0.092, -0.006] in A with the proxy, an interval that excludes zero, but the two sides differ in more than the interactions: X_{int} is built from the meal kernels, which $R_{\text{ref}}+L_{\text{lab}}$ does not contain, and adding the kernels alone to R_{ref} changes the error by -0.060 [-0.109, -0.021] in the same setting. This does not show that the change comes from the kernels alone, nor that the interactions are without value; the contribution of the interactions is judged from the main hierarchical comparison, which already controls the kernels and the additive laboratory states (Table 29). The differences between C-S1 and C-S2 in Table 29 are small and in the same direction; the two constructions span a similar linear space when the kernels are present, but the ridge penalty is not invariant to the reparametrization, so predictions need not be identical.

Table 31: E11: ΔRMSE [95% CI] of adding X_{int} to R_{ref}+K_{meal}+L_{lab} by meal stratum (mg/dL; number of participants with at least 20 windows in the stratum in parentheses): P30 = a meal in the last 30 minutes, P31–120 = last meal 31–120 minutes before, NoMeal120 = no meal in 120 minutes. The strata are descriptive; participants and RMSE are not additive across strata.

Setting	P30	P31–120	NoMeal120
A, C-S1	−0.116 [−0.222, −0.010] (44)	+0.003 [−0.099, +0.087] (44)	−0.006 [−0.012, +0.001] (44)
A, C-S2	−0.097 [−0.209, +0.018] (44)	+0.002 [−0.100, +0.085] (44)	−0.007 [−0.012, −0.001] (44)
A, C-S1, proxy	−0.113 [−0.230, +0.003] (44)	+0.005 [−0.101, +0.089] (44)	−0.008 [−0.013, −0.003] (44)
A, C-S2, proxy	−0.092 [−0.222, +0.039] (44)	+0.007 [−0.097, +0.092] (44)	−0.009 [−0.014, −0.004] (44)
B, C-S1	−0.467 [−0.946, +0.019] (8)	+0.113 [−0.112, +0.358] (8)	+0.006 [−0.019, +0.041] (9)
B, C-S2	−0.296 [−0.689, +0.060] (8)	+0.040 [−0.095, +0.194] (8)	−0.001 [−0.012, +0.012] (9)

Meal strata. The strata of Table 31 are those of Section 6.3; the 20-window rule applies in each stratum, so eight participants remain in B for P30 and P31–120 and nine for NoMeal120, and the penalty was not re-tuned per stratum. The larger point estimates shortly after a meal, the positive estimates later in the post-meal period and the intervals that depend on the construction and on the proxy do not support a stable post-meal interaction effect or an absorption mechanism. The overall change is not recovered by summing the strata, since the participants and the nonlinear RMSE differ.

Auxiliary selection. For C-S1, A without and with the proxy and B without the proxy, the twelve candidates were screened as for T (Section 4.5): subject-level Spearman correlation with the training residual on 15-minute down-sampled windows, the five candidates with the largest absolute mean correlation kept, the original redundancy filter at $|\rho| > 0.9$ against the 24 lags and the kept candidates, then forward selection on validation data with the paired one-standard-error rule, starting from R_{ref}+K_{meal}+L_{lab}. All twelve had a valid screening statistic; the five retained candidates were two breakfast interactions (fasting glucose, HbA1c) and three dinner interactions (HbA1c, HOMA-IR, fasting glucose), with a different order in the two protocols, and lunch and snack interactions did not enter the forward pool under this rule. In A, both settings selected the three dinner interactions with HbA1c, HOMA-IR and fasting glucose (`lab_dinner_carbs_x_hba1c_percent`, `lab_dinner_carbs_x_homa_ir`, `lab_dinner_carbs_x_fasting_glucose`); in B the paired rule selected none, although the forward curve had its minimum at four interactions, and the unpaired rule selected none in any setting (Table 32). The selected subset in A has an interval that barely excludes zero, but the subset and the penalty were chosen on the same validation data, so the interval ignores selection uncertainty. The main analysis remains the fixed twelve-interaction block.

Table 32: E11: auxiliary screening and forward selection among the twelve interactions (C-S1; paired one-standard-error rule on validation data). The selected subset and the penalty were chosen on the same validation data used for the evaluation. In B no interaction was selected, so both models are identical and the zero difference is not an estimate of a biological null effect.

Setting	Selected	Without	With selected X	ΔRMSE [95% CI]
A	3	10.796	10.782	−0.0135 [−0.0275, −0.0002]
A, proxy	3	10.772	10.757	−0.0143 [−0.0300, −0.0003]
B	0	12.188	12.188	0.0000 [0.0000, 0.0000]

Limits. Beyond the general limits of the follow-up analyses (Sections 6.5 and 9), Protocol B has nine participants (eight in two strata), the laboratory results have unknown return times so the analysis is offline, the original M features retain the activity-filter and interpolation availability limits of Section 3.3, no full-cohort sensitivity was run, and the observed associations are predictive associations under one fixed linear baseline and one Ridge representation, not physiological interactions.

G Supplementary Result Tables and Figures

This appendix collects the complete tables and figures that the main text summarizes or cites: the visual summary of the primary and secondary comparisons across validation, grouped cross-validation and test (Figure 5), the core and full cohorts (Table 33), the stable factors (Table 34), the comparison with the original paper (Table 35), the results by HbA1c group (Table 36) and the dose–response of the stable factors (Figure 6).

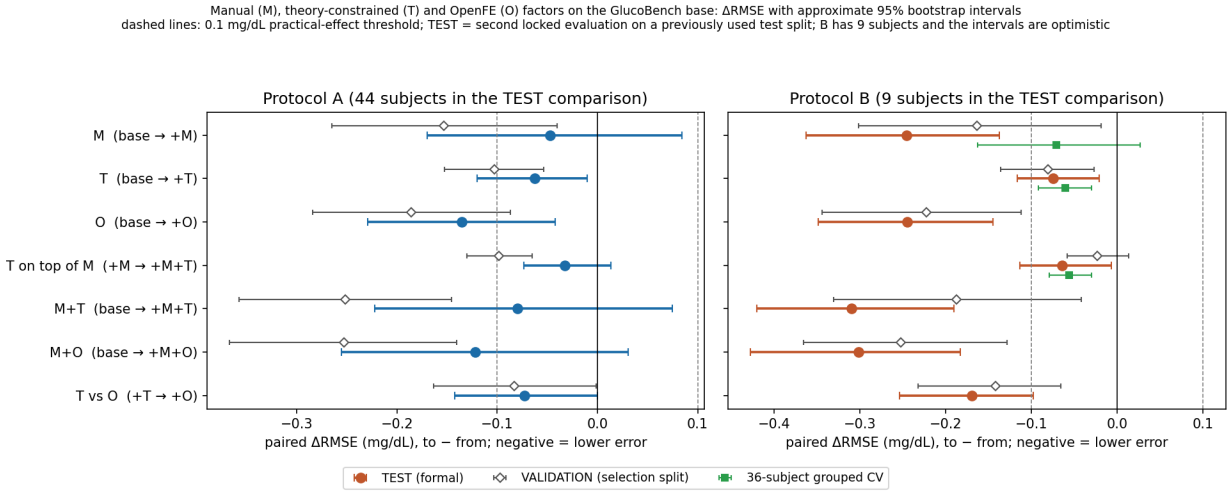


Figure 5: ΔRMSE of manual (M), theory-constrained (T) and OpenFE (O) factor blocks on the GlucoBench base with approximate 95% bootstrap intervals (negative = lower error). TEST uses a previously used test split (Section 6.4); VALIDATION is the selection split and optimistic; the 36-subject grouped CV exists for Protocol B only. Dashed lines: 0.1 mg/dL practical threshold. Protocol B has nine test subjects.

Figure 5 summarizes the primary and secondary contrasts of Table 5 at the three evidence stages. Negative values favor the added block; the test points are those of Table 5, and the other two stages show how the estimates change between the selection stages and the test split.

Table 33: Core versus full test cohort (supplementary; full-cohort values are outside the multiplicity family): every primary (P1–P4), secondary (S1–S3) and E1 contrast of Section 7.2 on both cohorts.

ID	Comparison	Core ΔRMSE	Core 95% CI	Full ΔRMSE	Full 95% CI	Impr. (full)
Protocol A: core $n = 44$, full $n = 45$						
P1	base → base+M	−0.047	[−0.170, +0.084]	+0.326	[+0.132, +0.542]	31%
P2	base → base+T	−0.063	[−0.120, −0.010]	−0.055	[−0.113, −0.005]	60%
P3	base+M → base+M+T	−0.033	[−0.073, +0.013]	−0.037	[−0.073, +0.004]	76%
P4	base → base+O	−0.135	[−0.229, −0.042]	−0.124	[−0.207, −0.043]	62%
S1	base → base+M+T	−0.080	[−0.222, +0.075]	+0.288	[+0.087, +0.513]	38%
S2	base → base+M+O	−0.122	[−0.256, +0.031]	+0.220	[+0.029, +0.430]	42%
S3	base+T → base+O	−0.073	[−0.143, −0.0002]	−0.069	[−0.128, −0.011]	64%
E1-1	Base → Base+A	+0.026	[−0.045, +0.126]	+0.360	[+0.187, +0.558]	27%
E1-6	Base+D+B → Base+A+D+B	+0.025	[−0.049, +0.125]	+0.403	[+0.227, +0.603]	31%
Protocol B: core $n = 9$, full $n = 9$						
P1	base → base+M	−0.245	[−0.362, −0.137]	−0.236	[−0.352, −0.132]	89%
P2	base → base+T	−0.075	[−0.116, −0.021]	−0.077	[−0.119, −0.022]	78%
P3	base+M → base+M+T	−0.064	[−0.114, −0.007]	−0.066	[−0.115, −0.010]	78%
P4	base → base+O	−0.244	[−0.349, −0.145]	−0.240	[−0.342, −0.143]	100%
S1	base → base+M+T	−0.309	[−0.420, −0.190]	−0.302	[−0.407, −0.185]	89%
S2	base → base+M+O	−0.301	[−0.427, −0.183]	−0.296	[−0.421, −0.177]	100%
S3	base+T → base+O	−0.169	[−0.253, −0.098]	−0.163	[−0.246, −0.094]	100%
E1-1	Base → Base+A	−0.017	[−0.048, +0.011]	−0.015	[−0.043, +0.010]	56%
E1-6	Base+D+B → Base+A+D+B	−0.012	[−0.059, +0.032]	−0.009	[−0.053, +0.033]	56%

Table 33 sets the core-cohort contrasts beside the same contrasts on the full cohort. In Protocol A the contrasts that introduce the manual set or the activity block differ between the two cohorts (the reversal

discussed in Section 7.2), whereas the contrasts that introduce only T or O keep their direction.

Table 34: Most frequently selected candidate factors (E4/E7): membership of the selected set T_k, stability at threshold 0.6 (A/B), half-sample selection frequency over 50 complementary pairs and leave-one-subject-out inclusion rate (LOSO). Source = earliest source minute before the origin. Frequencies are descriptive, not error rates.

Factor	Family	Source (min)	T_k A/B	Stable A/B	Freq. A/B	LOSO A/B
native_last	native_glucose_state	14	no/no	yes/yes	1.00/0.76	1.00/1.00
actstat_calories_max_15m	activity_kernels	15	no/no	yes/yes	0.94/0.84	1.00/1.00
ratio_last_meal_carb_share	nutrient_ratios	120	no/yes	yes/no	0.88/0.28	1.00/0.19
native_slope_1h	native_glucose_state	74	no/no	yes/yes	0.75/0.73	1.00/0.96
actstat_calories_mean_15m	activity_kernels	15	yes/no	yes/yes	0.68/0.66	0.96/0.93
native_diff	native_glucose_state	29	no/no	yes/yes	0.68/0.68	0.96/0.93
int_lunch_carbs_x_bmi	meal_x_baseline_state	120	yes/no	yes/no	0.60/0.49	0.89/0.67
int_snack_carbs_x_native_slope_1h	meal_x_baseline_state	120	yes/yes	no/yes	0.49/0.88	0.78/0.96
actstat_hr_mean_120m	activity_kernels	120	yes/no	no/no	0.37/0.27	0.42/0.04
actstat_hr_max_15m	activity_kernels	15	yes/yes	no/no	0.23/0.34	0.13/0.00
int_breakfast_carbs_x_native_slope_1h	meal_x_baseline_state	120	yes/no	no/no	0.32/0.33	0.24/0.04
int_dinner_carbs_x_native_slope_1h	meal_x_baseline_state	120	no/yes	no/yes	0.50/0.84	0.38/1.00
abs_lunch_carbs_t20_k2	meal_absorption	120	no/yes	no/no	0.12/0.10	0.18/0.37
ratio_kernel_protein_x_carbs	nutrient_ratios	120	no/yes	no/no	0.34/0.48	0.13/0.56
abs_dinner_fiber_t20_k3	meal_absorption	120	no/yes	no/no	0.11/0.01	0.07/0.00

Table 34 lists the candidates most often selected under subsampling, with their membership of T_k and their leave-one-subject-out inclusion; a stable factor need not belong to T_k (Section 7.5).

Table 35 compares the predictors of the original study with this project's evidence, qualitatively.

Table 35: Directional comparison with the original paper's in-sample SHAP findings (author-reported) and the evidence of this study; the agreement column is our own judgment.

Original paper (SHAP, in-sample)	This project	Agreement
Carbohydrate positive (iAUC 2nd, AUC 3rd)	Manual diet block in B: E1 contrasts about −0.20; largest gain in the P30 stratum; in A diet small and interval includes zero; the T set, which includes carbohydrate kernels, −0.063 (A) and −0.075 (B)	consistent in B; weak in A
Protein negative, fat weakly negative	Nutrient-ratio family: removing it raises B error by +0.023; the logged fat in the last 30 minutes adds −0.025 in A (single-feature addition to the base, validation); signs not tested	weak; not tested
Fasting glucose, HbA1c (iAUC 1st/5th)	Additive laboratory values: −0.078 in A on validation, −0.0002 after adding a subject-identity proxy; the meal-by-laboratory-state interactions gave no clear overall validation gain in E11, an offline predictive analysis that cannot confirm SHAP-based mechanisms	cannot be compared
Meal-start (baseline) glucose strongest for AUC	the last inferred native reading is stable in both protocols (frequency 1.0 in A, 0.76 in B); the base already contains glucose history	consistent in direction

Table 36 splits the primary test comparisons by HbA1c group (descriptive; Appendix F).

Table 36: Test ΔRMSE of the primary comparisons by HbA1c group (core cohort; descriptive). Groups have 14–16 participants in A and three in B.

ID	Comparison	HbA1c < 5.7	5.7–6.4	> 6.4
A:P1	base → base+M	+0.097 (n=14)	−0.063 (n=16)	−0.174 (n=14)
A:P2	base → base+T	−0.074 (n=14)	−0.071 (n=16)	−0.042 (n=14)
A:P3	base+M → base+M+T	−0.043 (n=14)	−0.056 (n=16)	+0.004 (n=14)
A:P4	base → base+O	−0.074 (n=14)	−0.237 (n=16)	−0.081 (n=14)
B:P1	base → base+M	−0.149 (n=3)	−0.340 (n=3)	−0.246 (n=3)
B:P2	base → base+T	−0.038 (n=3)	−0.119 (n=3)	−0.068 (n=3)
B:P3	base+M → base+M+T	−0.108 (n=3)	−0.083 (n=3)	−0.002 (n=3)
B:P4	base → base+O	−0.191 (n=3)	−0.303 (n=3)	−0.239 (n=3)

Dose–response of the stable factors (training windows). Figure 6 describes the association of the stable factors with the baseline residual *in training windows only*. Mean residual increases with activity calories over 15 minutes (from about −0.8 to −1.5 at the lowest deciles to about +1.1 to +1.4 mg/dL at the highest) and rises steeply with the lunch carbohydrate by BMI interaction at its top deciles (about +6 to +8 mg/dL, few windows, wide intervals). These are in-sample associations in training windows: they show that recent activity energy and the lunch carbohydrate by BMI term co-vary with what the baseline misses, not why, and they are not mechanisms.

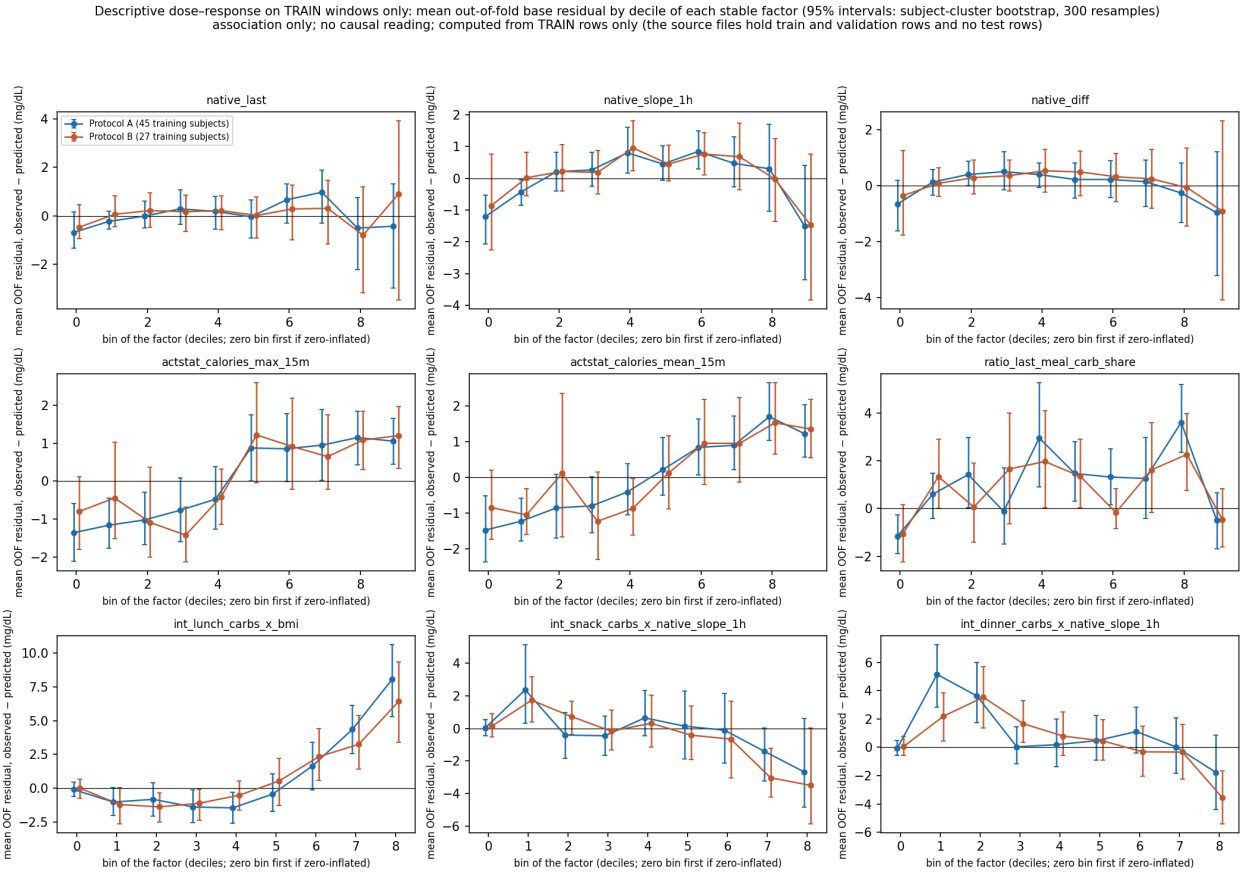


Figure 6: Descriptive dose–response of the stable factors on TRAIN windows only (mean out-of-fold base residual by decile; 95% subject-cluster bootstrap intervals, 300 resamples). Association only; no causal reading.

H Complete Formal Comparisons and Cohort Coverage

H.1 All 50 Formal Test Comparisons

Table 37 lists all 50 formal comparisons, including those with adverse or null results. Core cohort with the 20-window rule; the last column gives the validation ΔRMSE and the full-cohort test ΔRMSE (with the number of full-cohort participants). Practical-effect labels are I/N/W classes written out; p and Holm are exploratory approximate bootstrap p-values.

Table 37: All 50 formal test comparisons (core cohort, 20-window rule; "Full" = full cohort, supplementary). ΔRMSE = to − from, mg/dL; negative = lower error for the second configuration. p and Holm are exploratory approximate bootstrap p-values; only P1–P4 have a Holm value. VAL = VALIDATION ΔRMSE (selection-stage, optimistic). Label: practical-effect classification.

ID	Comparison	n	RMSE from→to	ΔRMSE	95% CI	Impr.	p	Holm	Label	VAL / Full ΔRMSE (full n)
A:P1	base → base+M	44	10.957→10.910	−0.047	[−0.170, +0.084]	52%	0.462	0.462	negligible	−0.153 / +0.326 (45)
A:P2	base → base+T	44	10.957→10.895	−0.063	[−0.120, −0.010]	64%	0.024	0.072	negligible	−0.103 / −0.055 (45)
A:P3	base+M → base+M+T	44	10.910→10.877	−0.033	[−0.073, +0.013]	64%	0.159	0.318	negligible	−0.098 / −0.037 (45)
A:P4	base → base+O	44	10.957→10.822	−0.135	[−0.229, −0.042]	59%	0.005	0.020	improved	−0.186 / −0.124 (45)
A:S1	base → base+M+T	44	10.957→10.877	−0.080	[−0.222, +0.075]	57%	0.306	–	negligible	−0.252 / +0.288 (45)
A:S2	base → base+M+O	44	10.957→10.835	−0.122	[−0.256, +0.031]	61%	0.113	–	improved	−0.253 / +0.220 (45)
A:S3	base+T → base+O	44	10.895→10.822	−0.073	[−0.143, −0.0002]	61%	0.051	–	negligible	−0.083 / −0.069 (45)
A:E1-1	Base → Base+A	44	10.957→10.983	+0.026	[−0.045, +0.126]	50%	0.669	–	negligible	−0.046 / +0.360 (45)
A:E1-2	Base+A → Base+A+D	44	10.983→10.932	−0.051	[−0.140, +0.029]	45%	0.221	–	negligible	−0.092 / −0.044 (45)
A:E1-3	Base+A+D → Base+A+D+B	44	10.932→10.910	−0.022	[−0.060, +0.016]	52%	0.274	–	negligible	−0.016 / +0.009 (45)
A:E1-4	Base → Base+D	44	10.957→10.904	−0.053	[−0.144, +0.026]	50%	0.208	–	negligible	−0.108 / −0.056 (45)
A:E1-5	Base → Base+B	44	10.957→10.939	−0.018	[−0.053, +0.017]	57%	0.335	–	negligible	−0.013 / −0.019 (45)
A:E1-6	Base+D+B → Base+A+D+B	44	10.885→10.910	+0.025	[−0.049, +0.125]	59%	0.709	–	negligible	−0.035 / +0.403 (45)
A:E1-7	Base+A+B → Base+A+D+B	44	10.962→10.910	−0.052	[−0.141, +0.029]	45%	0.219	–	negligible	−0.089 / −0.045 (45)
A:E5-1	base+T → base+T −activity_kernels	44	10.895→10.939	+0.044	[+0.006, +0.083]	30%	0.022	–	negligible	+0.067 / +0.036 (45)
A:E5-2	base+T → base+T −meal_x_baseline_state	44	10.895→10.907	+0.012	[−0.029, +0.057]	45%	0.615	–	negligible	+0.038 / +0.013 (45)
A:E9-1	base+M@logged → base+M@delay_free	44	10.910→11.344	+0.434	[+0.227, +0.710]	27%	0.001	–	worse	+0.084 / +0.195 (45)
A:E9-2	base+M@logged → base+M@delay_all	44	10.910→10.970	+0.061	[−0.012, +0.141]	55%	0.097	–	negligible	+0.101 / +0.045 (45)
A:E9-3	base+T@logged → base+T@delay_free	44	10.895→10.891	−0.004	[−0.013, +0.003]	48%	0.434	–	negligible	+0.009 / −0.003 (45)
A:E9-4	base+T@logged → base+T@delay_all	44	10.895→10.900	+0.005	[−0.021, +0.033]	41%	0.716	–	negligible	+0.021 / +0.004 (45)
A:E8-1	original base, reliable windows → anchor-causal base	43	10.897→15.060	+4.163	[+3.771, +4.584]	0%	0.001	–	worse	+4.229 / +4.101 (44)
A:E8-2	anchor-causal base → anchor-causal base+M	43	15.060→14.780	−0.280	[−0.498, −0.062]	56%	0.017	–	improved	−0.505 / +0.249 (44)

ID	Comparison	n	RMSE from→to	ΔRMSE	95% CI	Impr.	p	Holm	Label	VAL / Full ΔRMSE (full n)
A:E8-3	anchor-causal base → anchor-causal base+T	43	15.060→14.945	−0.115	[−0.199, −0.033]	65%	0.004	–	improved	−0.205 / −0.100 (44)
A:E8-4	anchor-causal base → anchor-causal base+O	43	15.060→14.761	−0.299	[−0.500, −0.079]	63%	0.009	–	improved	−0.453 / −0.276 (44)
B:P1	base → base+M	9	11.327→11.081	−0.245	[−0.362, −0.137]	89%	0.001	0.004	improved	−0.164 / −0.236 (9)
B:P2	base → base+T	9	11.327→11.252	−0.075	[−0.116, −0.021]	78%	0.015	0.030	negligible	−0.080 / −0.077 (9)
B:P3	base+M → base+M+T	9	11.081→11.017	−0.064	[−0.114, −0.007]	78%	0.033	0.033	negligible	−0.023 / −0.066 (9)
B:P4	base → base+O	9	11.327→11.082	−0.244	[−0.349, −0.145]	100%	0.001	0.004	improved	−0.222 / −0.240 (9)
B:S1	base → base+M+T	9	11.327→11.017	−0.309	[−0.420, −0.190]	89%	0.001	–	improved	−0.187 / −0.302 (9)
B:S2	base → base+M+O	9	11.327→11.025	−0.301	[−0.427, −0.183]	100%	0.001	–	improved	−0.252 / −0.296 (9)
B:S3	base+T → base+O	9	11.252→11.082	−0.169	[−0.253, −0.098]	100%	0.001	–	improved	−0.142 / −0.163 (9)
B:E1-1	Base → Base+A	9	11.327→11.310	−0.017	[−0.048, +0.011]	56%	0.246	–	negligible	−0.032 / −0.015 (9)
B:E1-2	Base+A → Base+A+D	9	11.310→11.109	−0.200	[−0.324, −0.086]	78%	0.001	–	improved	−0.123 / −0.193 (9)
B:E1-3	Base+A+D → Base+A+D+B	9	11.109→11.081	−0.028	[−0.064, −0.002]	89%	0.032	–	negligible	−0.009 / −0.028 (9)
B:E1-4	Base → Base+D	9	11.327→11.122	−0.204	[−0.345, −0.079]	89%	0.001	–	improved	−0.133 / −0.197 (9)
B:E1-5	Base → Base+B	9	11.327→11.304	−0.023	[−0.051, +0.001]	44%	0.063	–	negligible	+0.017 / −0.025 (9)
B:E1-6	Base+D+B → Base+A+D+B	9	11.094→11.081	−0.012	[−0.059, +0.032]	56%	0.580	–	negligible	−0.044 / −0.009 (9)
B:E1-7	Base+A+B → Base+A+D+B	9	11.284→11.081	−0.202	[−0.331, −0.086]	78%	0.001	–	improved	−0.120 / −0.195 (9)
B:E5-1	base+T → base+T −meal_absorption	9	11.252→11.261	+0.009	[−0.006, +0.024]	44%	0.262	–	negligible	+0.013 / +0.008 (9)
B:E5-2	base+T → base+T −nutrient_ratios	9	11.252→11.274	+0.023	[+0.005, +0.043]	22%	0.012	–	negligible	+0.012 / +0.022 (9)
B:E5-3	base+T → base+T −activity_kernels	9	11.252→11.244	−0.008	[−0.038, +0.019]	44%	0.643	–	negligible	+0.027 / −0.008 (9)
B:E5-4	base+T → base+T −meal_x_baseline_state	9	11.252→11.284	+0.032	[+0.007, +0.055]	11%	0.015	–	negligible	+0.008 / +0.037 (9)
B:E9-1	base+M@logged → base+M@delay_free	9	11.081→11.085	+0.003	[−0.045, +0.059]	44%	0.943	–	negligible	−0.004 / +0.005 (9)
B:E9-2	base+M@logged → base+M@delay_all	9	11.081→11.273	+0.192	[+0.089, +0.307]	11%	0.001	–	worse	+0.100 / +0.186 (9)
B:E9-3	base+T@logged → base+T@delay_free	9	11.252→11.273	+0.021	[+0.011, +0.033]	11%	0.001	–	negligible	+0.019 / +0.021 (9)
B:E9-4	base+T@logged → base+T@delay_all	9	11.252→11.321	+0.069	[+0.035, +0.101]	11%	0.001	–	negligible	+0.048 / +0.066 (9)
B:E8-1	original base, reliable windows → anchor-causal base	9	11.283→15.392	+4.109	[+3.587, +4.701]	0%	0.001	–	worse	+4.794 / +4.087 (9)
B:E8-2	anchor-causal base → anchor-causal base+M	9	15.392→14.653	−0.739	[−0.989, −0.497]	100%	0.001	–	improved	−0.720 / −0.710 (9)
B:E8-3	anchor-causal base → anchor-causal base+T	9	15.392→15.136	−0.256	[−0.373, −0.119]	89%	0.001	–	improved	−0.309 / −0.244 (9)
B:E8-4	anchor-causal base → anchor-causal base+O	9	15.392→14.836	−0.556	[−0.751, −0.400]	100%	0.001	–	improved	−0.519 / −0.533 (9)

H.2 Cohort Coverage

Table 38: Windows and participants by protocol, split and cohort. "Lost" = admitted windows outside the core cohort.

Protocol	Split	Cohort	Windows	Admitted	Lost	Subjects
A	train	core	79,300	79,300	0	45 of 45
A	train	full	79,300	79,300	0	45 of 45
A	validation	core	26,301	26,330	29	45 of 45
A	validation	full	26,330	26,330	0	45 of 45
A	test	core	21,083	26,269	5,186	44 of 45
A	test	full	26,269	26,269	0	45 of 45
B	train	core	79,294	81,958	2,664	27 of 27
B	train	full	81,958	81,958	0	27 of 27
B	validation	core	23,139	24,259	1,120	9 of 9
B	validation	full	24,259	24,259	0	9 of 9
B	test	core	26,861	28,322	1,461	9 of 9
B	test	full	28,322	28,322	0	9 of 9